

Johns Hopkins University
EN.605.755 Systems Biology
Instructor: Dr. Christopher Bradburne

A Ras-Driven Cancer Stem Cell–Microenvironment Circuit Provides a Mechanistic Basis for Malignant State Stability

AJ Book & Shrinithi Natarajan

Due: December 9, 2025

Contents

Abstract	2
1 Background, Approach, and Results	2
2 Discussion	43
3 Conclusions	45
4 Individual Team Member Contributions	46
References	47

Abstract

Cutaneous squamous cell carcinoma (cSCC) in HRAS^{G12V} mouse skin progresses from benign papillomas to invasive lesions as cancer stem cells (CSCs) co-evolve with an angiogenic, TGF β -rich niche and hijack leptin–LEPR–mTOR signalling. Building on Yuan *et al.*, we asked whether this Ras-initiated CSC–microenvironment circuit is dynamically sufficient to maintain a malignant state at fixed Ras input.

We formalised the circuit in a six-variable ODE model tracking CSC fraction (C), angiogenesis (A), TGF β activity (T), LEPR competence (R), leptin availability (L), and PI3K–AKT–mTOR activity (M), with Ras as an external driver. Parameters were constrained by Fermi-style estimates and calibrated to z-scored RNA-seq and ATAC-seq modules for CSC, angiogenesis, TGF β , and mTOR across normal, papilloma, SCC, and Lepr^{-/-} tumours, then evaluated using steady-state analysis, Ras up/down continuation, and in silico perturbations.

The calibrated model admits two coexisting steady states under fixed Ras, a low- C /low- A benign-like basin and a high- C /high- A malignant basin, separated by an unstable saddle; Ras hysteresis yields a maximal CSC gap of 0.374 between branches. Weakening the leptin–LEPR–mTOR arm collapses the high-CSC attractor even with Ras held oncogenic, qualitatively recapitulating Lepr^{-/-} phenotypes, whereas anti-angiogenic therapy and Ras inhibition drive distinct residual CSC plateaus. Single-cell data show a bimodal CSC module distribution and strong CSC–mTOR coupling but weak TGF β –LEPR and LEPR–mTOR links, highlighting underconstrained edges. Overall, the results support a Ras-driven, feedback-stabilised mechanism for malignant stability and yield falsifiable predictions for vascular, TGF β , and leptin-pathway interventions.

1 Background, Approach, and Results

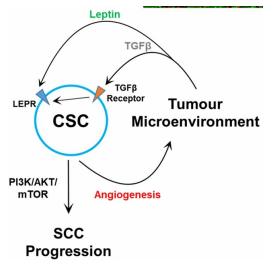
Background

Cutaneous squamous cell carcinoma (cSCC) arises in an epithelium that must constantly balance stem cell self-renewal, differentiation, and mechanical stress. Lineage tracing and single-cell analyses in multistage mouse skin carcinogenesis show that oncogenic Ras does not simply expand a fixed cancer stem cell clone, but drives convergence into plastic stem-like states that trade off proliferation, multilineage potential, and drug tolerance [11, 18]. These malignant stem-like states evolve within a structurally complex tissue where stem cells actively shape and sense their niche through mechanical forces, phagocytic clearance of dying neighbours, and dynamic interactions with stromal and immune compartments [6, 17]. Broader work argues that tumour progression in skin and other epithelia reflects not only cumulative mutations but also non-genetic rewiring of stem cell–microenvironment circuits [24].

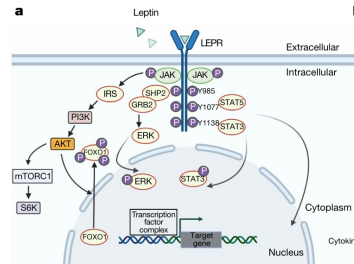
In a clonal HRAS^{G12V} mouse model of skin carcinogenesis, Yuan and colleagues showed that cancer stem cells progressively rewire a circuit with the surrounding microenvironment [23]. As lesions progress from benign papillomas to invasive cSCC, these cells induce angiogenesis, stimulate TGF β signalling at the tumour–stroma interface, and eventually hijack leptin and leptin receptor (LEPR) signalling to activate PI3K–AKT–mTOR during the benign-to-malignant transition [23]. Mosaic skin studies further show that Ras-mutant clones are

constrained by wild-type neighbours: after injury, EGFR-driven proliferation in wild-type stem cells can outcompete Ras-mutant cells, indicating that the balance between mutant and wild-type stem cells is dynamically set by local repair and signalling context [7]. In parallel, work in mouse and human models defines how TGF β signalling, tumour-associated fibroblasts, and perivascular mesenchymal progenitors remodel vasculature and impose long-lived invasive and drug-resistant phenotypes on tumour stem cells [1, 2, 14, 22, 25], while VEGF-driven niches support epidermal cancer stem cell survival and the formation of highly vascularized tumours [9, 10]. Together, these data support viewing Ras-driven cSCC as a distributed network linking cancer stem cells, vasculature, and TGF β -rich niches, rather than as a purely cell-autonomous consequence of Ras activation.

Leptin and LEPR provide a metabolic and inflammatory axis into this network. Beyond its canonical role in energy balance, leptin–LEPR signalling has been implicated in proliferation, inflammation, and metastasis across cancers, with activity in tumour, stromal, and immune cells [4, 13]. High systemic leptin levels, characteristic of obesity, correlate with enhanced tumour growth and altered immune composition in several settings [13]. Mechanistically, leptin activates PI3K and mTOR pathways to drive vascular smooth muscle proliferation and neointimal hyperplasia, and hyperleptinaemia increases the dose of mTOR inhibition required to curb pathological growth; combined PI3K plus mTOR blockade is more effective than mTOR inhibition alone [16]. In the HRAS^{G12V} skin tumour model, cancer stem cells eventually co-opt leptin–LEPR signalling to secure PI3K–AKT–mTOR activation during malignant progression [23]. This positions leptin–LEPR–mTOR as a candidate control node within a Ras–CSC–microenvironment circuit.



A CSC–microenvironment circuit cartoon (Yuan *et al.*, Ext. Fig. 8G).



B Leptin–LEPR signalling cascade (Yuan *et al.*, Fig. 5A).

Figure 1: Experimental schematics motivating the Ras–CSC–microenvironment model. Panel (A) shows how CSCs, leptin–LEPR, TGF β , and angiogenesis are wired in HRAS-driven skin tumours, and panel (B) shows the molecular leptin–LEPR pathway that we coarse-grain into an $R, L \rightarrow M \rightarrow C$ axis.

Taken together, these strands motivate treating Ras-driven cSCC as a feedback system in which oncogenic Ras, cancer stem cell fraction, angiogenesis, TGF β signalling, and leptin–LEPR–mTOR form a coupled circuit embedded in an active niche. In the next step we formalise this circuit in a coarse-grained model, define the variables C, A, T, R, M , and state the hypotheses we will test.

Step 1: Define the hypothesis and biological question

Building on the experimental sequence described above, we now recast the Yuan et al. findings as a concrete dynamical question. In the HRAS^{G12V} skin tumour model, lesions progress from benign papillomas to invasive cSCC as cancer stem cells and their niche co-evolve: CSCs induce angiogenesis, stimulate TGF β signalling at the tumour–stroma interface, and ultimately hijack leptin–LEPR signalling to engage PI3K–AKT–mTOR. This trajectory yields a malignant state that is enriched for CSCs, highly angiogenic, and more drug-resistant, without any further increase in oncogenic Ras. The original work documents these transitions at the level of lineage tracing, single-cell and bulk RNA-seq, chromatin, and functional perturbations, but it does not ask whether the Ras–CSC–microenvironment circuit, by itself, is dynamically sufficient to generate and maintain a distinct malignant state.

To make this question precise, we formalize a minimal representation of the circuit by aggregating the key components into six coarse-grained variables: CSC fraction C , angiogenesis A , TGF β activity T , LEPR expression and signalling competence R , local leptin availability L , and leptin–mTOR pathway activity M . Oncogenic Ras is treated as a persistent input set by the initiating mutation and not directly manipulated in the model. The experimental sequence in Yuan et al. can then be read as Ras increasing C ; CSCs and associated stromal cells increasing A and T ; TGF β and vascular niches promoting LEPR-high CSC states (R); angiogenesis enhancing local leptin availability L ; and leptin–LEPR signalling raising M to further expand or stabilise C . This motivates the positive feedback chain

$$C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C,$$

with oncogenic Ras providing a persistent upstream drive into both C and A , embedded in the mechanically and immunologically active tissue context described in the Background.

The biological question for a hypothesis-driven model is therefore sharpened as follows. If Ras is held constitutively active, and if other pathways are held at baseline, is this Ras-initiated CSC–microenvironment feedback loop by itself able to support two qualitatively distinct regimes, one with high CSC fraction and high angiogenesis and one with low CSC fraction and low angiogenesis, that behave as separate steady states of the system? Furthermore, does targeted disruption of the leptin–LEPR–mTOR arm of the loop, through genetic or pharmacologic reduction of R or M , suffice to collapse a high- C , high- A malignant state back toward a low- C , low- A state even when Ras remains active?

In this project the model is explicitly constructed to test that hypothesis. We treat Ras as an external, fixed driver, and we encode the Ras–CSC–microenvironment circuit in a small set of coupled dynamical variables C, A, T, R, L, M . We then ask whether, for biologically plausible parameter values, the resulting system admits (i) a high- C , high- A steady state that corresponds to the malignant, leptin-hijacked configuration observed in Yuan et al., and (ii) a low- C , low- A steady state that corresponds to a benign or partially suppressed configuration. We also ask whether partial or complete inhibition of LEPR or mTOR (reducing R or M in the model) produces a qualitative shift from the high- C , high- A branch to the low- C , low- A branch, as would be required if leptin–LEPR–mTOR is a genuine control point rather than a passive reporter of malignant progression.

Formally, we state the hypotheses in terms of steady states and interventions on R and M .

Alternative hypothesis H_1 . In Ras-driven skin tumours, the positive feedback loop

$$C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C$$

is sufficient to generate and maintain a high-CSC, highly angiogenic malignant state. In the dynamical model this means that, for fixed oncogenic Ras input and biologically plausible parameters, there exist at least two stable steady states for (C, A, T, R, L, M) : a high- C , high- A state and a low- C , low- A state. Under this hypothesis, genetic or pharmacologic disruption of LEPR or mTOR signalling, modelled as a sustained reduction in R or M , eliminates the high- C , high- A steady state or makes it unstable, and trajectories that start in the malignant basin move toward a low- C , low- A attractor even though Ras remains active.

Null hypothesis H_0 . The Ras-initiated CSC–microenvironment feedback loop mediated by A , T , R , L , and M is not sufficient to generate or maintain a distinct high-CSC, high-angiogenesis malignant state. In the model, either there is only a single stable steady state for (C, A, T, R, L, M) for fixed Ras, or any high- C , high- A regime that appears is not destabilised by reducing R or M . In particular, under H_0 , perturbations that inhibit LEPR or mTOR signalling do not produce a qualitative shift from a high- C , high- A state to a low- C , low- A state; at most they cause small quantitative changes in CSC fraction and angiogenesis that do not correspond to leaving one attractor basin and entering another.

The goal of the subsequent steps is to build an explicit Ras–CSC–microenvironment model that is simple enough to analyse yet faithful to the sequence reported by Yuan et al., then to determine which of these hypotheses is supported by the behaviour of the model over a realistic parameter range.

Step 2: Prior knowledge and data landscape

Mechanistic studies in skin and related epithelia provide a detailed picture of how TGF β , vasculature, and leptin–LEPR interact with oncogenic signalling in exactly the way a Ras-initiated feedback loop would require. Conditional deletion of TGF β receptors in LGR5⁺ stem cells cooperates with MAPK pathway hyperactivation or TP53 loss to accelerate cSCC formation, implicating loss of canonical TGF β –SMAD signalling as a key event in tumour initiation [2]. Organotypic human skin models show that loss of TGF β RII, which disables both SMAD4-dependent and SMAD4-independent branches, is sufficient to induce invasive growth and to rewire adhesion and MAPK signalling networks [22]. Within established SCCs, TGF β activity is spatially heterogeneous: TGF β -responding tumour stem cells positioned near vasculature adopt slower-cycling, invasive, and drug-resistant phenotypes via p21–NRF2–glutathione programs, whereas non-responding progeny proliferate faster and form the tumour bulk [14]. These findings support a picture in which TGF β -rich vascular niches impose long-lived phenotypic states on cancer stem cells, providing one arm of the positive feedback that our hypothesis requires if a malignant CSC-rich state is to be self-sustaining under fixed Ras input.

TGF β also feeds back on the vasculature and stromal compartments in a way that naturally fits into the $C \rightarrow A \rightarrow T$ segment of the Ras–CSC–microenvironment circuit. Tumour-associated fibroblasts promote microvessel organization and lumen size by increasing

TGF β signalling, fibronectin deposition, and pericyte coverage of endothelial tubes, generating more stable yet tumour-supportive vessels [25]. In mesenchymal progenitors, p38 α restrains a TGF β - and JNK-driven angiogenic program that includes mesenchymal–endothelial transition; loss of p38 α amplifies angiogenesis and tumour growth in colon cancer models [1]. In epidermal tumours, VEGF signalling provides an additional coupling between cancer stem cells and the vascular niche: autocrine VEGF cooperates with EGFR to drive proliferation in skin cancer models [10], and a subset of epidermal cancer stem cells depends on VEGF-A–neuropilin-1 signalling for survival, invasion, and formation of highly vascularized tumours [9]. Together, these data motivate modelling angiogenesis as both an output of cancer stem cell activity and an input that feeds back on CSC behaviour through TGF β - and VEGF-driven niches—exactly the structure needed if C and A are to share a high-activity malignant branch distinct from a benign branch, as posited under H_1 .

Leptin and LEPR connect this circuit to systemic metabolism and immune tone and supply the specific control arm we intend to test. Beyond its canonical role in energy balance, leptin has been implicated in multiple hallmarks of cancer, including proliferation, inflammation, and metastasis, with leptin–LEPR signalling reported in tumour, stromal, and immune compartments [4, 13]. Hyperleptinaemia correlates with enhanced tumour growth and altered immune composition in several settings. Mechanistically, leptin activates PI3K and mTOR pathways in multiple cell types and increases the dose of mTOR inhibition required to curb pathological growth; combined PI3K + mTOR blockade is more effective than mTOR inhibition alone [16]. In the HRAS^{G12V} skin tumour model, cancer stem cells eventually co-opt leptin–LEPR signalling during the benign-to-malignant transition to secure sustained PI3K–AKT–mTOR activity [23]. In our modelling framework, these observations justify treating local leptin availability $L(t)$ and downstream PI3K–AKT–mTOR activity $M(t)$ as a distinct axis in the circuit and motivate the specific hypothesis that the $R, L \rightarrow M \rightarrow C$ branch is stabilising rather than merely descriptive of malignant progression; under the null hypothesis, weakening this branch would produce only small, graded changes in C and A rather than a shift between attractors.

From a systems perspective, this biology maps naturally onto small signalling motifs known to generate multistability and hysteresis, which we treat as prior structural evidence in favour of H_1 . Synthetic and natural gene circuits show that positive feedback and mutual activation can function as bistable switches and memory modules [8, 21]. Multisite phosphorylation and kinase cascades can generate ultrasensitive and bistable responses even without explicit feedback [12, 19]. Ultrasensitive modules embedded in feedback create robust switches and oscillators [3, 5], and dual positive-feedback motifs enlarge the parameter regimes that support bimodal responses and history dependence [20]. Analyses of autoregulation and network topology demonstrate how such motifs increase the number and stability of steady states in multistable fate circuits [15]. In the context of Ras-driven cSCC, this theory provides prior justification for aggregating Ras activity, CSC fraction, angiogenesis, TGF β signalling, LEPR competence, leptin availability, and mTOR signalling into a coarse-grained positive-feedback module (C, A, T, R, L, M) that is capable, in principle, of supporting both benign and malignant attractors.

The data landscape for this project then determines whether this hypothesised feedback structure is even compatible with observations. Single-cell RNA-seq from the HRAS^{G12V} model provides state-resolved expression patterns for Ras signatures, stemness programs,

angiogenic factors, $TGF\beta$ targets, and leptin–mTOR pathway genes across benign papilloma and malignant cSCC stages. Within the $K14^+$ tumour-cell compartment, the transcriptional state space forms a small number of communities in UMAP space (Fig. 2). A compact CSC-enriched island occupies a restricted region, and *Lepr* expression is concentrated in a hotspot that partially overlaps this island. Cluster-level summaries show that only a subset of $K14^+$ clusters are jointly Krt14-high and *Lepr*-high. These observations support two modelling choices directly tied to the hypothesis: treating $C(t)$ as a CSC-enriched subpopulation within the broader $K14^+$ compartment, and representing $R(t)$ as a fractional LEPR competence within that CSC pool, enabling the $R, L \rightarrow M \rightarrow C$ branch to select and stabilise a malignant subpopulation.

Complementary chromatin data (ATAC-seq) capture accessibility at motifs associated with the same pathways (for example SMADs, AP-1, TEAD, STATs, NF- κ B, FOXO), providing an independent readout of pathway engagement and justifying the inclusion of distinct $TGF\beta$ and mTOR axes. Spatial measurements localize $TGF\beta$ activity, vessel density, and LEPR expression to perivascular niches where invasive and drug-resistant CSCs reside, matching the spatial pattern expected if a high- C , high- A attractor is realised in $TGF\beta$ -rich vascular regions. Finally, bulk RNA-seq across normal, papilloma, SCC, PDV_WT, and PDV_LeprKO lesions can be summarised into module activities for $TGF\beta$, mTOR, angiogenesis, and CSC programmes (Fig. 3). For this hypothesis-based model we use these measurements in a coarse way: as constraints on which aggregate variables should appear, how they should be coupled, and what qualitative separations (benign vs. malignant, WT vs. *Lepr*^{-/-}) the model must reproduce. Under H_1 there should exist parameter regimes in which a single dynamical system on (C, A, T, R, L, M) can place these conditions on distinct regions of a benign vs. malignant branch; under H_0 no such regime exists, or the effect of *Lepr*^{-/-} cannot be recapitulated without adding additional mechanisms. These considerations define the modelling constraints used in subsequent steps, where we formalise a coarse-grained feedback model and test whether realistic parameter values can produce a self-sustaining high-CSC malignant state and predict the effect of perturbing the leptin–mTOR arm.

Step 3: Formulate a mental model

To translate the hypothesis from Step 1 into a concrete circuit, we represent the Ras-driven CSC–microenvironment system as a small network of six coarse-grained state variables, with Ras treated as an external input fixed by the initiating mutation. The nodes are: cancer stem cell fraction $C(t)$, angiogenesis $A(t)$, active $TGF\beta$ level $T(t)$, LEPR expression in CSCs $R(t)$, local leptin concentration $L(t)$, and PI3K–AKT–mTOR activity $M(t)$. Each node summarises a cluster of molecular and cellular processes, but is defined operationally in terms of measurable quantities (such as CSC fraction from single-cell RNA-seq, CD31⁺ vessel density, $TGF\beta$ reporter activity, LEPR staining, leptin ELISA, and pS6/pAKT levels). In other words, the six variables (C, A, T, R, L, M) are the axes along which H_1 and H_0 will be evaluated.

The mental model is that oncogenic Ras supplies a constant driving input that seeds both a low basal CSC fraction and a modest pro-angiogenic state. CSCs in turn are modelled as the main amplifiers of angiogenesis, via secretion of VEGFA and related factors, so that the CSC fraction C and vessel density A co-evolve. Angiogenesis raises local $TGF\beta$ concentrations,

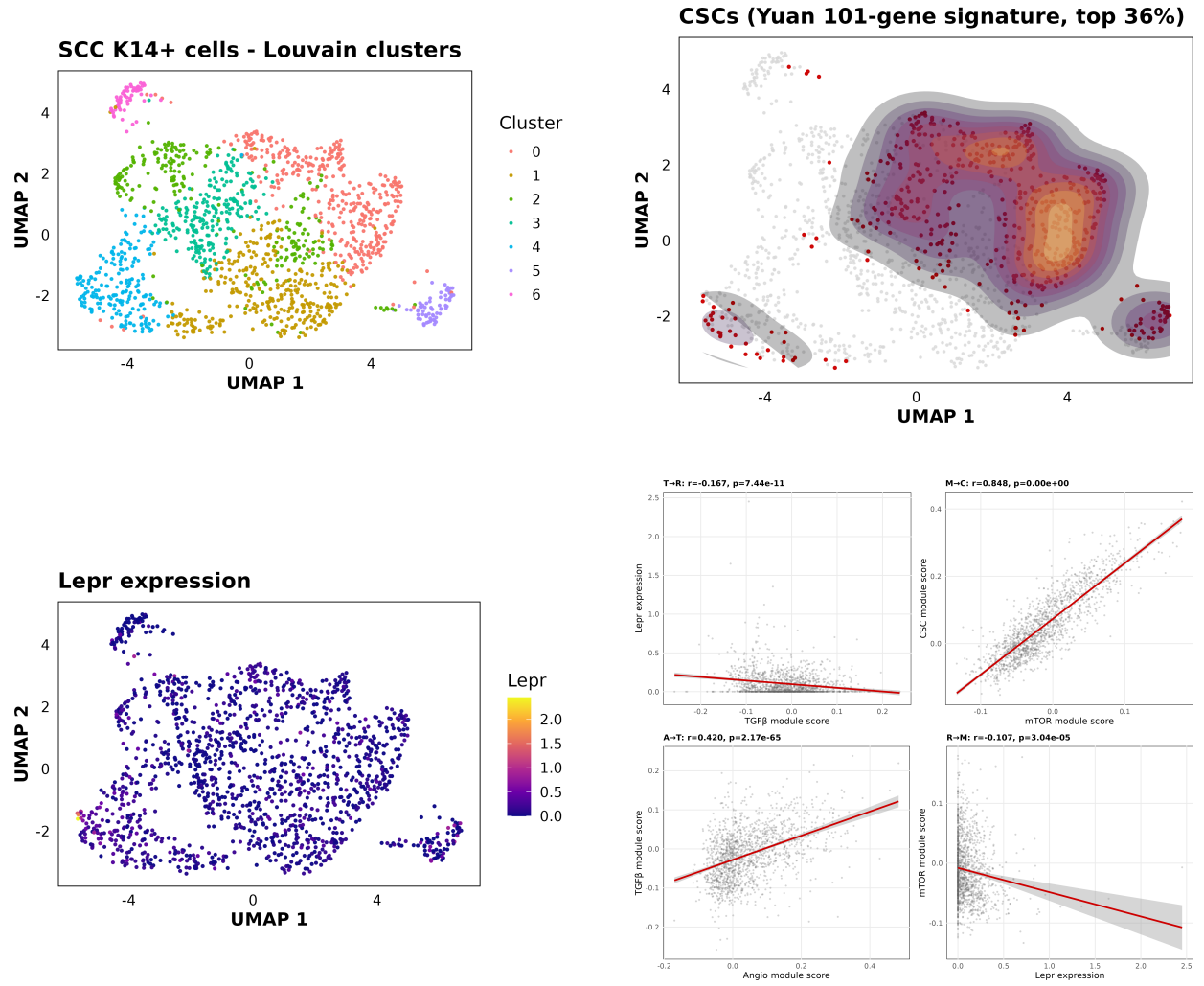


Figure 2: Single-cell SCC landscape and constraints on the Ras-CSC-microenvironment circuit. Top: Louvain clusters of K14⁺ SCC tumour cells (left) define the transcriptional state space, and a CSC-enriched island identified by the Yuan 101-gene CSC signature occupies a restricted region (right). Bottom: *Lepr* expression forms a hotspot that only partially overlaps the CSC island (left), and a mechanistic correlation panel (right) summarises empirical tests of the $T \rightarrow R$, $R \rightarrow M$, $M \rightarrow C$, and $A \rightarrow T$ edges.

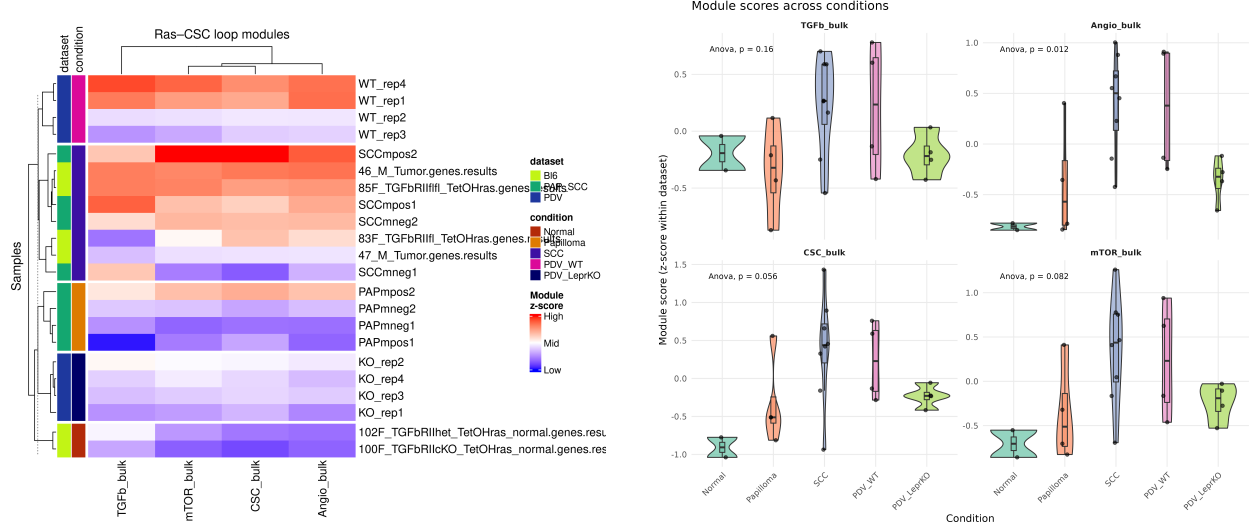


Figure 3: Bulk RNA-seq summaries used to define module targets for the Ras-CSC-microenvironment model. Left: sample-level heatmap of TGF β , mTOR, angiogenesis, and CSC module scores across BI6, PAP_SCC, and PDV series, annotated by dataset and condition, showing coordinated co-variation with lesion stage and *Lepr* status. Right: condition-level module score distributions (z-scores) with ANOVA p -values, defining the calibration targets (C_i^{data} , A_i^{data} , T_i^{data} , M_i^{data}) used in Step 6.

primarily by recruiting and activating perivascular stromal and immune cells, with an optional contribution of CSC-derived TGF β that can be tuned from negligible to moderate depending on the data. TGF β induces LEPR in CSCs in a switch-like fashion, creating a LEPR-high CSC subpopulation described by $R(t)$. Angiogenesis also increases local leptin availability $L(t)$ by importing systemic leptin from the circulation into the tumour microenvironment. Leptin and LEPR together drive PI3K-AKT-mTOR activity $M(t)$ in CSCs. Finally, TGF β and mTOR signalling feed back to increase the CSC fraction C by converting non-CSCs into CSCs and by supporting CSC survival and proliferation.

This wiring implements the positive feedback loop

$$C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C,$$

with Ras acting as an external driver into both C and A , and with an optional additional input from C to T . At the level of a mental model, the hypothesis H_1 is now a statement about the global behaviour of this loop: for fixed Ras input, the closed circuit should be able to support at least two internally consistent configurations, one with low C and low A (benign, papilloma-like) and one with high C and high A (malignant, SCC-like), and it should expose the leptin-LEPR-mTOR arm (R and M) as a control handle whose sustained inhibition can collapse the high- C , high- A configuration even when Ras remains on. The null hypothesis H_0 corresponds to the same wiring diagram failing to show such behaviour: the loop either admits only a single stable configuration at fixed Ras, or any high- C , high- A regime that appears cannot be eliminated by reducing R or M .

In subsequent steps, each arrow in this mental model will be translated into an explicit term in a set of ODEs or quasi-steady relations, and the resulting dynamical system will

be probed to see whether it realises the bistability and control properties required by H_1 or instead behaves more like H_0 .

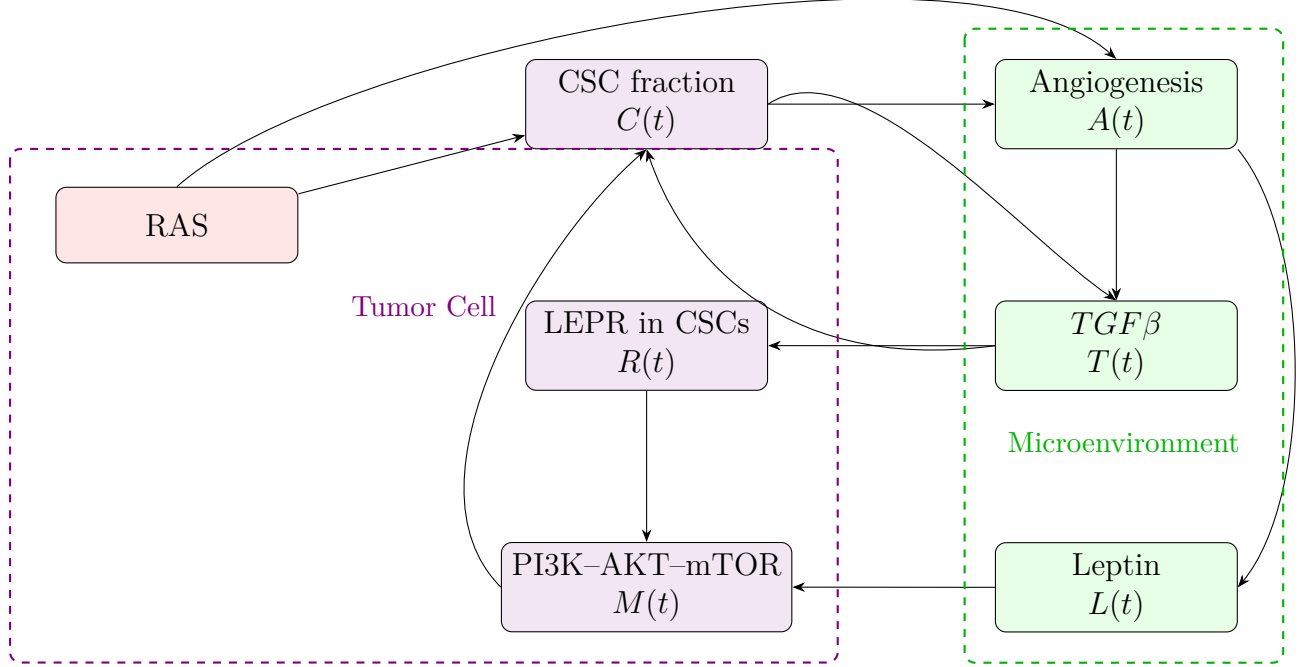


Figure 4: Mental-model diagram of the Ras–CSC–microenvironment feedback circuit, which encodes the working hypothesis H_1 . Ras provides a constant oncogenic input into both CSC seeding and pro-angiogenic drive. CSCs amplify angiogenesis ($C \rightarrow A$); angiogenesis and CSC burden increase local $TGF\beta$ ($A, C \rightarrow T$); $TGF\beta$ induces LEPR in CSCs ($T \rightarrow R$); angiogenesis raises leptin availability ($A \rightarrow L$); leptin and LEPR together activate PI3K–AKT–mTOR ($L, R \rightarrow M$); and mTOR activity feeds back to expand the CSC fraction ($M \rightarrow C$), closing the loop. Under H_1 , this closed circuit at fixed Ras admits both benign-like and malignant-like steady states and exposes the leptin–LEPR–mTOR arm as a control lever; under H_0 , the same wiring fails to show those properties.

Step 4: Decide on the modelling concept

Given the hypothesis and mental model, we adopt a small, continuous-time, network-based model in which the six coarse-grained variables

$$C(t), A(t), T(t), R(t), L(t), M(t)$$

evolve according to deterministic ordinary differential equations (ODEs). The central question in H_1 and H_0 is about the existence, stability, and controllability of steady states at fixed Ras input, not about cell-to-cell noise or spatial patterning per se. A low-dimensional ODE system is therefore the minimal framework that can (i) encode the Ras–CSC–microenvironment

circuit from Step 3 and (ii) allow direct tests of whether the loop admits multiple stable steady states and whether perturbing R or M can collapse a high- C , high- A attractor.

Each variable is treated as a dimensionless activity or fraction scaled to the interval $[0, 1]$, so that the model tracks relative levels (for example, “low”, “intermediate”, “high” CSC fraction or angiogenesis) rather than absolute molecule counts. This scaling is chosen to match the nature of the calibration data in Steps 6–7, which are z-scored module activities, and to keep the phase space bounded for the steady-state analysis that will be used to test H_1 vs H_0 . Oncogenic Ras is not given its own state variable; instead it enters as a fixed input parameter that continuously drives a basal CSC fraction and a basal angiogenic signal, reflecting the clonal HRAS^{G12V} mutation in the mouse model and aligning with the hypothesis that Ras is held “on” while the feedback loop determines whether the system sits in a benign-like or malignant-like attractor.

Conceptually, the network is written as a set of production-minus-loss balances for each node in the feedback loop

$$C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C,$$

with additional Ras inputs into C and A . Production terms encode how upstream variables increase a downstream node, while loss terms summarise decay, deactivation, or dilution. To represent the fact that many of these interactions are graded but switch-like, we use smooth sigmoidal input–output functions (Hill-type nonlinearities) rather than purely linear activation. In practice this means that, for example, TGF β activity T increases LEPR competence R slowly at low T , more steeply around a threshold, and then approaches a plateau at high T . Moderate cooperativity (Hill coefficients of order two to four) is allowed so that the circuit can, in principle, generate multistable behaviour and hysteresis over Ras or feedback gains, which is exactly what H_1 claims the motif should be able to do.

Timescale separation is used in a minimal way, only where it matters for the attractor structure. Variables that reflect relatively fast biochemical or diffusional processes, such as local leptin availability L or short-lived TGF β activity T , are treated as quickly relaxing toward quasi-steady levels set by slower variables (CSC fraction and angiogenesis). In contrast, the CSC-related quantities C, R, M and the vascular variable A are treated as slow and explicitly dynamic, because they involve changes in cell composition, vessel remodelling, and stable transcriptional programs. This separation keeps the effective dimensionality low while preserving the delays and feedback structure needed for bistability in the Ras–CSC–microenvironment loop and for the hysteresis and “memory” experiments that will later be used to test whether the malignant state is a genuine attractor.

Stochasticity is not modelled explicitly at the level of the circuit. The primary data motivating this project (stage-stratified single-cell and bulk measurements across lesions) constrain mean behaviours and coarse-grained module activities rather than individual stochastic trajectories. Randomness is therefore represented implicitly through parameter uncertainty and variation in initial conditions, rather than through explicit noise terms in the ODEs. This choice keeps the model analytically tractable for steady-state, bifurcation, and sensitivity analysis, which are the tools needed to ask whether the wiring in Fig. 4 is sufficient to support distinct benign-like and malignant-like steady states under fixed Ras input and whether targeted reductions in R or M can shift trajectories from one basin of attraction to another.

Alternative modelling paradigms such as spatially explicit reaction–diffusion equations, agent-based simulations that track individual cells, or fully stochastic chemical master equation descriptions would be more detailed but would introduce many additional parameters that are not well constrained by the available data and would make it harder to interpret failures of H_1 in terms of the core feedback structure. For this hypothesis-driven project we therefore prioritise a low-dimensional, deterministic, network-based ODE formulation that directly mirrors the wiring in Figure 4 and is designed specifically to probe the existence, stability, and controllability of high- C , high- A versus low- C , low- A regimes within a single Ras-driven feedback circuit.

Step 5: Formulate the first simple mathematical model

The annotated circuit in Fig. 5 is taken as the blueprint for a first explicit mathematical model. Each node in the diagram is represented by a scalar state variable

$$C(t), A(t), T(t), R(t), L(t), M(t) \in [0, 1],$$

which denote, respectively, the CSC fraction, angiogenesis, TGF β activity, LEPR competence in CSCs, local leptin availability, and PI3K–AKT–mTOR activity. The Ras node is not assigned its own dynamical equation and instead appears as a fixed, dimensionless input $f_{\text{RAS}} \in [0, 1]$ that quantifies the strength of the clonal HRAS^{G12V} drive. All state variables are rescaled to the interval $[0, 1]$, so the model tracks relative activity levels rather than absolute molecule counts or cell numbers. This matches the data that will later be used for calibration, which enter as z-scored module activities rather than absolute copy numbers.

The equations are constructed by translating each arrow in Fig. 5 into a production term for the downstream node and subtracting a first-order loss term. For a generic variable $X(t)$, the balance has the form

$$\frac{dX}{dt} = (\text{sum of production terms from incoming arrows}) - \delta_X X, \quad (1)$$

where $\delta_X > 0$ is a decay, deactivation, or loss rate. Whenever the biological interaction is graded but saturating and exhibits a threshold-like onset, the corresponding arrow is represented by a Hill activation function. For a non-negative input x , the Hill function is written as

$$H_n(x; K) = \frac{x^n}{K^n + x^n}, \quad (2)$$

where $K > 0$ sets the half-maximal point and $n \geq 1$ controls the steepness. This form rises smoothly from 0 to 1 as x increases and allows induction to be weak at low input, steep near a threshold, and saturating at high input, which matches the qualitative behaviour expected for TGF β -dependent LEPR induction and leptin–LEPR-dependent mTOR activation.

The CSC fraction $C(t)$ sits at the top of the loop and accumulates three distinct inputs in Fig. 5. The Ras $\rightarrow C$ arrow carries a basal seeding term $\rho_C f_{\text{RAS}}$, representing the continuous low-level conversion of non-CSCs into CSCs driven directly by oncogenic Ras. The $T \rightarrow C$ arrow carries a TGF β -dependent conversion term $\eta_C H_n(T; K_C)$, reflecting the experimentally observed promotion of invasive, stem-like phenotypes in high-TGF β niches. The $M \rightarrow C$ arrow contributes an mTOR-dependent support term $\eta_{C,M} M$, summarising the ability of

PI3K–AKT–mTOR signalling to promote CSC survival and self-renewal. All of these processes act on the pool of non-CSCs, which is proportional to $1 - C$, so the combined production term is multiplied by $1 - C$. CSCs are lost by differentiation or death at rate δ_C . The resulting balance for the CSC fraction is

$$\frac{dC}{dt} = \left[\rho_C f_{\text{RAS}} + \eta_C H_n(T; K_C) + \eta_{C,M} M \right] (1 - C) - \delta_C C. \quad (3)$$

This equation makes the $\text{Ras} \rightarrow C$, $T \rightarrow C$, and $M \rightarrow C$ arrows explicit. Ras continuously seeds a basal CSC pool, TGF β converts non-CSCs into CSCs in a thresholded fashion through the Hill term $H_n(T; K_C)$, and mTOR activity further supports expansion of the CSC compartment. The factor $1 - C$ ensures that CSC production vanishes as C approaches its upper bound.

The angiogenesis variable $A(t)$ integrates two inputs in the diagram: a Ras-driven pro-angiogenic signal and a CSC-driven vascular amplification. The $\text{Ras} \rightarrow A$ arc in Fig. 5 is labelled $\alpha_A f_{\text{RAS}}$, representing direct Ras-related induction of angiogenic factors in the epithelium and its microenvironment. The $C \rightarrow A$ arrow is labelled $\beta_A C$ and captures the secretion of VEGF and other pro-angiogenic mediators by CSCs. Both contributions act to increase vessel density, but angiogenesis cannot grow indefinitely; the remaining capacity is proportional to $1 - A$. Vessels regress or remodel at rate δ_A . The angiogenesis balance is therefore

$$\frac{dA}{dt} = \left[\alpha_A f_{\text{RAS}} + \beta_A C \right] (1 - A) - \delta_A A. \quad (4)$$

Here the $\text{Ras} \rightarrow A$ arrow becomes the term $\alpha_A f_{\text{RAS}}(1 - A)$, and the $C \rightarrow A$ arrow becomes $\beta_A C(1 - A)$. When Ras or CSCs are active and A is low, angiogenesis grows; as A approaches one, these production terms are automatically suppressed.

TGF β activity $T(t)$ sits downstream of both angiogenesis and CSC burden. The arrow from A to T is labelled $\alpha_T A$, reflecting the observation that TGF β -rich perivascular niches emerge as vasculature develops. The arrow from C to T is labelled $\gamma_T C$ and represents CSC-derived TGF β production, which can be tuned to be small or moderate. At this level of coarse graining, TGF β activity is treated as a linear function of these two sources, with a first-order decay rate δ_T , giving

$$\frac{dT}{dt} = \alpha_T A + \gamma_T C - \delta_T T. \quad (5)$$

Because both A and C are themselves bounded between 0 and 1, this linear form is sufficient for a first model; nonlinearity is already present elsewhere in the loop through the Hill terms, and adding another saturation here would not change the qualitative structure we are testing in H_1 versus H_0 .

LEPR competence $R(t)$ is a fractional property of the CSC compartment. In the diagram, the $T \rightarrow R$ arrow is annotated with $\eta_R T^p / (K_R^p + T^p)$, indicating a thresholded induction of LEPR in response to TGF β . To ensure that only LEPR-low CSCs can be converted, this term is multiplied by the remaining LEPR-negative fraction $1 - R$. LEPR competence can be lost or downregulated with rate δ_R . The resulting equation is

$$\frac{dR}{dt} = \eta_R H_p(T; K_R) (1 - R) - \delta_R R, \quad (6)$$

where $H_p(T; K_R)$ is the Hill function in Eq. (2) with exponent p and half-maximal point K_R . This equation makes the $T \rightarrow R$ arrow precise: $\text{TGF}\beta$ imposes a switch-like increase in the fraction of CSCs that are LEPR-competent, and that process slows as R approaches one.

Local leptin availability $L(t)$ is linked directly to angiogenesis in Fig. 5. The arrow from A to L is annotated by the quasi-steady relationship $(\mu_L/\delta_L)A$, which is the steady-state solution of a simple production–loss balance in which leptin import or production is proportional to vessel density and leptin is cleared with rate δ_L . Making this underlying dynamics explicit yields

$$\frac{dL}{dt} = \mu_L A - \delta_L L. \quad (7)$$

If A is held fixed, Eq. (7) relaxes to $L^* = (\mu_L/\delta_L)A$, matching the label on the arrow in the figure. In the full system, L tracks changes in A on a timescale set by $1/\delta_L$, which corresponds to the equilibration of leptin levels in the vascularised microenvironment.

The final node, mTOR activity $M(t)$, integrates leptin and LEPR competence. The two arrows entering M in Fig. 5 are summarised by a Hill function of the effective signal LR , written below the node as

$$F_M(L, R) = \eta_M \frac{(LR)^q}{K_M^q + (LR)^q}, \quad (8)$$

where η_M scales the maximal mTOR activity, K_M sets the sensitivity to combined leptin–LEPR signalling, and q controls the sharpness of the activation. The product LR represents the effective LEP–LEPR signalling strength, since both ligand availability and receptor competence are required. As with the other fractional variables, mTOR activity cannot exceed 1, so the production term is multiplied by $1 - M$ and a first-order loss term $\delta_M M$ is subtracted. This gives

$$\frac{dM}{dt} = F_M(L, R)(1 - M) - \delta_M M = \eta_M \frac{(LR)^q}{K_M^q + (LR)^q} (1 - M) - \delta_M M. \quad (9)$$

This equation implements the converging $L \rightarrow M$ and $R \rightarrow M$ arrows: mTOR is activated only when both leptin and LEPR are present at sufficient levels, and this activation saturates as the product LR increases.

Equations (3)–(9), together with the generic balance in Eq. (1) and the Hill activation in Eq. (2), define a six-dimensional ODE system for the Ras-driven CSC–microenvironment circuit. Every term has a direct origin in the annotated wiring of Fig. 5. Ras input appears only in the $\text{Ras} \rightarrow C$ and $\text{Ras} \rightarrow A$ terms; angiogenesis and CSC burden jointly control $\text{TGF}\beta$; $\text{TGF}\beta$ sets the pace of LEPR induction; angiogenesis controls leptin; leptin and LEPR together set mTOR activity; and mTOR and $\text{TGF}\beta$ both feed back to increase the CSC fraction. This is therefore the minimal deterministic ODE realisation of the mental model in Step 3, with nonlinearities introduced only at biologically motivated switch-like steps. In the next steps we use this system to ask whether, for fixed Ras input, the circuit can support both low- C , low- A and high- C , high- A steady states and whether interventions on R or M can collapse the malignant state as hypothesised in H_1 .

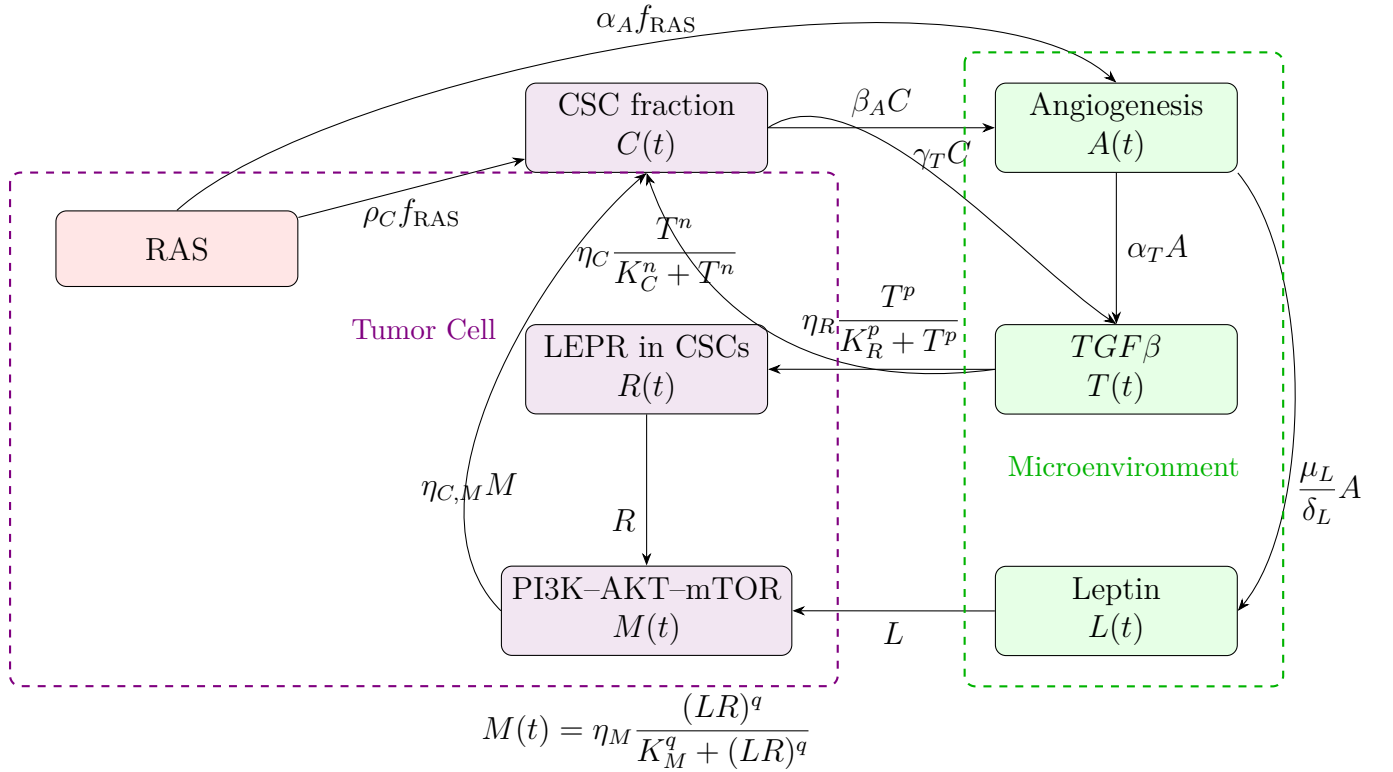


Figure 5: Annotated Ras–CSC–microenvironment feedback circuit with the main coupling terms from Eqs. (3)–(9) shown on the arrows.

Step 6: Test the model performance against available data

Having specified the Ras–CSC–microenvironment circuit in Eqs. (3)–(9), the first question is whether a single parameter set can reproduce the coarse pattern of pathway activity seen across lesion classes in the Yuan *et al.* experiments. The goal is not full parameter identification but a sanity check aligned with the hypothesis: does a six-variable circuit with fixed Ras input live in the same qualitative regime as the bulk RNA-seq and ATAC-seq data, or is it fundamentally inconsistent with the multistage progression and the *Lepr* perturbation?

Construction of low-dimensional experimental targets

Bulk RNA-seq data from the Bl6, PAP_SCC, and PDV series were preprocessed into stage-averaged module scores. Four modules were defined per sample: a CSC programme, an angiogenesis programme, a TGF β target set, and a PI3K–mTOR signalling module. For each sample, genes within a module were variance-stabilised and averaged, and the resulting module activities were z-scored across all samples. These per-sample scores were then averaged over replicates within each dataset/lesion-class combination, yielding six RNA-based experimental conditions: Bl6 Normal, Bl6 SCC, PAP_SCC Papilloma, PAP_SCC SCC, PDV WT, and PDV *Lepr*^{−/−}.

In parallel, ATAC-seq data were summarised into accessibility-based analogues of the same four axes, using motif- and peak-level aggregation rather than expression. This produced an additional set of targets for HFSC, IFE, and papilloma or SCC subpopulations stratified by mTOR-activity status (mneg, mpos). Together, these RNA and ATAC summaries define a multi-omics calibration target for each condition on the four-dimensional space spanned by CSC activity C , angiogenesis A , TGF β signalling T , and mTOR signalling M .

For each condition i the RNA-based targets can be written as

$$\mathbf{y}_i^{\text{data}} = (C_i^{\text{data}}, A_i^{\text{data}}, T_i^{\text{data}}, M_i^{\text{data}}), \quad (10)$$

where each component is a z-scored module summary. Higher values of C_i^{data} , A_i^{data} , T_i^{data} , and M_i^{data} correspond to stronger CSC, angiogenic, TGF β , and mTOR activity in the underlying lesions. The combined RNA and ATAC targets are summarised in Table 1.

Table 1: Multi-omics calibration targets for CSC (C), angiogenesis (A), TGF β (T), and mTOR (M). RNA entries are mean (SE) in z -score space; ATAC entries are mean accessibility-based targets without standard errors.

Modality	Dataset/Sample	Condition	C	A	T	M
RNA	Bl6	Normal	-0.907 (0.132)	-0.816 (0.035)	-0.193 (0.152)	-0.705 (0.154)
RNA	Bl6	SCC	0.453 (0.226)	0.408 (0.229)	0.096 (0.291)	0.352 (0.242)
RNA	PAP_SCC	Papilloma	-0.319 (0.301)	-0.396 (0.288)	-0.349 (0.207)	-0.360 (0.279)
RNA	PAP_SCC	SCC	0.319 (0.486)	0.396 (0.298)	0.349 (0.120)	0.360 (0.401)
RNA	PDV	PDV_LeprKO	-0.233 (0.074)	-0.355 (0.113)	-0.207 (0.095)	-0.235 (0.111)
RNA	PDV	PDV_WT	0.233 (0.259)	0.355 (0.316)	0.207 (0.288)	0.235 (0.328)
ATAC	HFSC	Normal	-0.513	-0.462	-0.063	-0.304
ATAC	IFE	Normal	0.466	0.480	0.601	0.652
ATAC	Papilloma_mneg	Papilloma	-0.064	-0.121	-0.294	-0.117
ATAC	Papilloma_mpos	Papilloma	-0.437	-0.246	-0.478	-0.499
ATAC	SCC_mneg	SCC	0.479	0.399	0.359	0.356
ATAC	SCC_mpos	SCC	0.069	-0.050	-0.125	-0.088

Numerically, the benign Bl6 normal samples sit at the low end of all four RNA axes; for example $C_{\text{Bl6,Normal}}^{\text{data}} = -0.907$, $A_{\text{Bl6,Normal}}^{\text{data}} = -0.816$, $T_{\text{Bl6,Normal}}^{\text{data}} = -0.193$, and $M_{\text{Bl6,Normal}}^{\text{data}} = -0.705$. Bl6 SCC and PAP_SCC SCC lesions lie at the high end ($C_{\text{Bl6,SCC}}^{\text{data}} = 0.453$, $A_{\text{Bl6,SCC}}^{\text{data}} = 0.408$, $T_{\text{Bl6,SCC}}^{\text{data}} = 0.096$, $M_{\text{Bl6,SCC}}^{\text{data}} = 0.352$; $C_{\text{PAP,SCC}}^{\text{data}} = 0.319$, $A_{\text{PAP,SCC}}^{\text{data}} = 0.396$, $T_{\text{PAP,SCC}}^{\text{data}} = 0.349$, $M_{\text{PAP,SCC}}^{\text{data}} = 0.360$). The PDV series shows a clear effect of *Lepr* status: PDV WT has positive CSC, angiogenesis, TGF β , and mTOR scores ($C_{\text{PDV,WT}}^{\text{data}} = 0.233$, $A_{\text{PDV,WT}}^{\text{data}} = 0.355$, $T_{\text{PDV,WT}}^{\text{data}} = 0.207$, $M_{\text{PDV,WT}}^{\text{data}} = 0.235$), whereas PDV *Lepr*^{-/-} shows approximately symmetric decreases ($C_{\text{PDV,LeprKO}}^{\text{data}} = -0.233$, $A_{\text{PDV,LeprKO}}^{\text{data}} = -0.355$, $T_{\text{PDV,LeprKO}}^{\text{data}} = -0.207$, $M_{\text{PDV,LeprKO}}^{\text{data}} = -0.235$). This symmetry makes the PDV pair a natural test of whether the $L, R \rightarrow M \rightarrow C$ arm behaves as a control lever (as required by H_1) or merely as a passive readout (as in H_0). The RNA-based targets and their confidence intervals are visualised in Fig. ??.

Mapping lesion classes to model inputs

To compare the model to these targets, each lesion class was mapped to a choice of Ras input $f_{\text{RAS},i}$ and, where appropriate, to an effective external TGF β drive. Normal skin was assigned

a low Ras input, papillomas an intermediate input, and cSCCs a higher value, consistent with sustained Ras-pathway activity along the benign-to-malignant axis. Numerically, the Ras input was set to $f_{\text{RAS}} = 0.3$ for normal-like conditions, $f_{\text{RAS}} = 0.6$ for papillomas, and $f_{\text{RAS}} = 0.95$ for SCC-like lesions. PDV WT and PDV *Lepr*^{-/-} were both treated as high-Ras conditions with $f_{\text{RAS}} = 0.95$, reflecting their derivation from an aggressive tumour model rather than from benign or pre-malignant lesions.

The Bl6 and PAP_SCC backgrounds were allowed modest shifts in an additive TGF β drive to reflect stromal and immune differences that are slowly varying compared to the model timescales. In contrast, PDV WT and PDV *Lepr*^{-/-} were assigned the same external TGF β drive and differed only in the strength of leptin-mTOR signalling.

The PDV *Lepr*^{-/-} class was treated as having the same Ras input and external TGF β drive as PDV WT but impaired leptin signalling, implemented as a reduction in the effective strength of the $L, R \rightarrow M$ term in Eq. (9). For each class i these choices define a fixed input vector

$$u_i = (f_{\text{RAS},i}, T_{\text{ext},i}), \quad (11)$$

which plays the role of condition-specific forcing in the ODE system. The same internal parameter vector θ (decay rates, Hill exponents, thresholds, and coupling strengths) is used for all classes, so that differences between lesions are attributed to Ras drive and niche context rather than to class-specific parameter sets. This is essential for the hypothesis: H_1 posits that a single Ras-initiated feedback circuit can realise both benign and malignant regimes under fixed Ras, whereas H_0 would effectively require different parameter regimes for different stages.

Model readout, normalisation, and error metric

Given a parameter vector θ and input u_i for class i , the six-dimensional system was integrated forward in time from benign-like initial conditions until the solution approached a steady state. Convergence was assessed by monitoring the norm of the time derivative and the change in the state vector over the final portion of the simulation. The resulting steady state

$$(C_i^*, A_i^*, T_i^*, R_i^*, L_i^*, M_i^*) \quad (12)$$

was then reduced to the same four-dimensional summary used on the data side:

$$\mathbf{y}_i^{\text{model}}(\theta) = (C_i^*, A_i^*, T_i^*, M_i^*). \quad (13)$$

LEPR competence R_i^* and leptin availability L_i^* do not have direct quantitative counterparts in the available bulk data and were therefore excluded from the calibration objective, although they are crucial for the perturbation and hysteresis experiments that follow.

Because the experimental module scores are z-scored, an affine transformation was applied to the model outputs along each axis so that the benign and malignant conditions span a comparable numerical range. This normalisation does not change the ordering of classes or the dependence on f_{RAS} ; it simply puts model and data into comparable units. The discrepancy between model and data for class i is captured by the residual vector

$$\mathbf{r}_i(\theta) = \mathbf{y}_i^{\text{model}}(\theta) - \mathbf{y}_i^{\text{data}}, \quad (14)$$

and the corresponding residual sum of squares

$$\text{RSS}(\theta) = \sum_i \|\mathbf{r}_i(\theta)\|_2^2. \quad (15)$$

Minimising this objective yields a data-informed parameter set $\hat{\theta}$.

Initial parameter priors and Fermi estimates

The optimisation is not started from arbitrary numerical seeds. Instead, the initial parameter vector is built from simple Fermi estimates based on the expected biological scale and relative influence of each process. These priors serve two roles: they constrain the search to a biologically reasonable region of parameter space and they demonstrate that the model can be brought into the correct qualitative regime starting from educated guesses rather than finely tuned values.

For the CSC equation, the basal Ras-driven seeding rate was set to $\rho_C \approx 0.15$, reflecting the expectation that direct Ras influence on stemness is present but modest, on the order of 10–20% of the maximal production rate. The mTOR-dependent amplification term was given a stronger prior weight, $\eta_{C,M} \approx 2.5$, to encode the idea that PI3K–AKT–mTOR activity substantially biases CSC self-renewal. The half-saturation constant for CSC induction was set to $K_C \approx 0.25$, corresponding to a low threshold at which CSCs begin to respond appreciably to upstream signals, and the Hill coefficient was set to $n_C \approx 3.5$ to represent a switch-like induction consistent with cooperative transcriptional control. The CSC decay rate was set to $\delta_C \approx 0.8$, implying a characteristic turnover time somewhat longer than the unit timescale used for the model but still fast enough to allow adaptation over lesion progression.

For angiogenesis, the Ras-dependent drive was assigned a moderate prior value $\alpha_A \approx 0.4$, corresponding to the view that Ras alone can initiate but not fully sustain a vascular programme. The CSC-dependent gain β_A was set near unity, $\beta_A \approx 1.0$, to reflect the strong experimental evidence that CSCs robustly secrete VEGF and other pro-angiogenic factors. The angiogenic decay rate δ_A was taken to be ≈ 0.6 , slower than the CSC decay rate, consistent with the idea that vessels remodel more slowly than individual tumour cells turn over.

For TGF β , the angiogenesis-dependent production coefficient was set to $\alpha_T \approx 0.8$, encoding the expectation that vascular niches bring substantial TGF β activity into the tumour microenvironment, whereas the CSC-dependent contribution γ_T was kept smaller, $\gamma_T \approx 0.15$, to represent a weaker direct secretion component. The TGF β decay rate was set to $\delta_T \approx 1.2$, in line with the relatively rapid turnover and clearance of this cytokine.

For LEPR induction, the TGF β -dependent gain was assigned $\eta_R \approx 0.6$, representing a moderate but non-negligible effect of TGF β on LEPR expression in CSCs. The half-saturation constant was set to $K_R \approx 0.35$, with a Hill coefficient $p_R \approx 2.5$, reflecting cooperative induction that is less steep than the CSC switch but still clearly nonlinear. LEPR decay was taken to be $\delta_R \approx 0.6$, comparable to the timescale of other transcriptionally regulated receptors.

For leptin availability, the angiogenesis-dependent production coefficient was set to $\mu_L \approx 0.9$, consistent with the idea that a well-developed vasculature can efficiently deliver circulating leptin into the tumour microenvironment, and the leptin clearance rate was taken as $\delta_L \approx 0.6$, reflecting moderate local equilibration dynamics.

For mTOR activation, the maximal activation strength was initially set to $\eta_M \approx 1.0$, with a half-saturation constant $K_M \approx 0.25$ for the effective leptin-LEPR signal LR , based on the notion that mTOR responds strongly to moderate upstream input. The Hill coefficient was chosen as $q_M \approx 3.5$ to model cooperative activation within the pathway, and the mTOR decay or relaxation rate was taken as $\delta_M \approx 0.6$.

These priors were complemented by biologically motivated bounds that restrict the search space during calibration. For example, the Ras-driven CSC seeding rate was allowed to vary within $0.05 \leq \rho_C \leq 0.5$, the mTOR-to-CSC gain within $0.5 \leq \eta_{C,M} \leq 5.0$, the CSC threshold within $0.1 \leq K_C \leq 0.5$, and the CSC Hill coefficient within $2.0 \leq n_C \leq 5.0$. Similar bounded ranges were imposed on angiogenesis, $TGF\beta$, and mTOR decay rates and gains. Ras inputs for each condition followed the normal-papilloma-SCC mapping described above. In combination, these Fermi estimates and bounds ensure that the optimisation explores a region of parameter space that is compatible with known timescales and relative effect sizes, rather than drifting into numerically convenient but biologically implausible regimes.

Calibration performance and residual structure

For the calibrated parameter set, the global residual sum of squares is

$$RSS(\hat{\theta}) = 10.23 \quad \text{over} \quad N = 24 \text{ data points,}$$

corresponding to a mean squared error of $MSE = 0.43$ and a root-mean-square deviation of

$$RMSE = \sqrt{MSE} \approx 0.65$$

in the same z-score units as the data. On average, the model reproduces each module score to within about two-thirds of a standard deviation. The module-wise contributions are summarised in Table 2. Angiogenesis shows the smallest residual sum of squares, followed by the CSC module and mTOR, while $TGF\beta$ exhibits the largest residual, consistent with the expectation that a single scalar $T(t)$ cannot fully capture spatially and temporally heterogeneous $TGF\beta$ niches.

Table 2: Calibration error summary across modules. RSS = residual sum of squares over all calibration targets; MSE = mean squared error.

Module	RSS	n	MSE
A	1.833	6	0.306
C	2.015	6	0.336
M	2.491	6	0.415
T	3.889	6	0.648
GLOBAL	10.228	24	0.426

The model-versus-data scatter plots (Fig. ??) show that the calibrated circuit preserves the experimental ordering across lesion classes: normal skin lies lowest on all four axes, papillomas occupy an intermediate regime, and SCC lesions (Bl6 and PAP_SCC) sit highest. PDV WT lies near the malignant SCC states, while PDV *Lepr*^{-/-} is shifted toward the

benign end along all axes. The residuals range from approximately -1.5 to $+0.8$ z-score units with a mean close to zero and an empirical standard deviation of about 0.66, and most residuals lie within ± 1 z-score unit. The detailed mapping from data to model values, including residuals for each module and condition, is reported in Table 3.

Table 3: Model fit to bulk RNA calibration targets in z -score space. Entries are data $\rightarrow$ model (residual), where residual = model – data.

Dataset	Condition	C	A	T	M
Bl6	Normal	$-0.907 \rightarrow -1.359 (-0.452)$	$-0.816 \rightarrow -1.752 (-0.935)$	$-0.193 \rightarrow -1.681 (-1.489)$	$-0.705 \rightarrow -1.013 (-0.308)$
Bl6	SCC	$0.453 \rightarrow 0.976 (0.523)$	$0.408 \rightarrow 0.873 (0.465)$	$0.096 \rightarrow 0.901 (0.805)$	$0.352 \rightarrow 1.000 (0.647)$
PAP_SCC	Papilloma	$-0.319 \rightarrow -0.964 (-0.645)$	$-0.396 \rightarrow -0.800 (-0.404)$	$-0.349 \rightarrow -0.840 (-0.490)$	$-0.360 \rightarrow -0.977 (-0.617)$
PAP_SCC	SCC	$0.319 \rightarrow 0.976 (0.657)$	$0.396 \rightarrow 0.873 (0.478)$	$0.349 \rightarrow 0.901 (0.551)$	$0.360 \rightarrow 1.000 (0.640)$
PDV	PDV_LeprKO	$-0.233 \rightarrow -0.606 (-0.373)$	$-0.355 \rightarrow -0.068 (0.287)$	$-0.207 \rightarrow -0.182 (0.026)$	$-0.235 \rightarrow -1.010 (-0.775)$
PDV	PDV_WT	$0.233 \rightarrow 0.976 (0.743)$	$0.355 \rightarrow 0.873 (0.518)$	$0.207 \rightarrow 0.901 (0.693)$	$0.235 \rightarrow 1.000 (0.765)$

Critically for the hypothesis, the PDV pair behaves in the expected direction. Under the mapping in Eq. (11), reducing the $L, R \rightarrow M$ term to mimic *Lepr* loss decreases the steady-state mTOR level and, via the $M \rightarrow C$ and $A \rightarrow T$ links, reduces CSC fraction, angiogenesis, and TGF β activity. The magnitude of the M shift between PDV WT and PDV *Lepr*^{-/-} is captured well (model: $\Delta M^* \approx -0.46$ versus data: -0.47), while the accompanying changes in C , A , and T are slightly smaller in the model than in the data. This pattern already suggests that the calibrated circuit routes a substantial share of positive feedback through the $L, R \rightarrow M \rightarrow C$ arm, as required under H_1 , but that a single TGF β variable cannot fully capture the measured spread in T .

The calibrated parameter values themselves are reported in Table 4. Most parameters remain within a factor of two of their initial Fermi estimates, which indicates that the prior reasoning about timescales and relative gains is compatible with the data, while a subset of parameters—particularly those governing TGF β production and decay—shift more substantially, reflecting the need to accommodate the observed spread in TGF β -related module scores across lesion classes.

Overall, this calibration demonstrates that a single Ras–CSC–microenvironment circuit with fixed Ras input can place all six RNA-based lesion classes along a benign-to-malignant continuum that resembles the experimental axis and can highlight *Lepr* status as a special control perturbation. The structured misfit, especially in the TGF β component, marks specific parts of the loop where additional biological detail (for example, multiple TGF β niches or explicit spatial segregation) would be needed for tight quantitative agreement.

Step 7: Refine the model and estimate parameters

The calibrated Ras–CSC–microenvironment circuit reproduces the coarse module patterns across lesion classes, but the hypothesis requires more than numerical agreement with bulk RNA targets. The refined model must also occupy a dynamical regime in which (i) the Ras input generates a genuine hysteresis loop in CSC burden and (ii) the leptin–LEPR–mTOR arm acts as a control axis whose sustained inhibition collapses a high- C malignant state. This section describes how the parameter vector $\hat{\theta}$ was refined and validated against these dynamical requirements and how local robustness was assessed.



Figure 6: Overview of model performance against RNA-seq calibration targets. Top: module targets (C, A, T, M) with 95% confidence intervals for each dataset and condition, defining the benign–papilloma–SCC and PDV WT versus *Lepr*^{-/-} structure that the model must reproduce. Bottom left: calibrated model steady states versus these targets for each condition. Bottom right: distribution of residuals (model – data) by module and condition.

Table 4: Calibrated parameters for the Ras–CSC feedback model. All parameters are dimensionless rate or gain coefficients unless otherwise noted.

Parameter	Value	Interpretation
ρ_C	0.191	CSC basal production.
η_C	1.200	Effect of Ras on CSC.
$\eta_{C,M}$	2.514	Effect of mTOR on CSC.
K_C	0.445	Ras half-saturation for CSC.
n_C	3.197	Ras Hill coefficient for CSC.
δ_C	1.172	CSC decay rate.
α_A	0.400	Basal angiogenesis drive.
β_A	1.220	CSC-driven angiogenesis gain.
δ_A	1.145	Angiogenesis decay rate.
α_T	0.985	Basal TGF β production.
γ_T	0.150	Angiogenesis influence on TGF β .
δ_T	1.578	TGF β decay rate.
η_R	0.600	TGF β influence on LEPR.
K_R	0.350	TGF β half-saturation for LEPR.
p_R	2.500	TGF β Hill coefficient for LEPR.
δ_R	0.600	LEPR decay rate.
μ_L	0.900	Angiogenesis-driven leptin delivery.
δ_L	0.600	Leptin decay rate.
η_M	1.157	Leptin–LEPR influence on mTOR.
K_M	0.250	Leptin–LEPR half-saturation for mTOR.
q_M	3.500	Leptin–LEPR Hill coefficient for mTOR.
δ_M	0.718	mTOR decay rate.

Dynamical refinement using Ras hysteresis and control perturbations

After fitting the module targets, the Ras input f_{RAS} was swept quasi-statically upward and then downward while tracking the steady-state CSC fraction. For each value of f_{RAS} , the ODE system was integrated to steady state from a low- C initial condition (benign-like) and from a high- C initial condition (malignant-like). The resulting branches define a Ras–CSC response curve that either collapses to a single line (no hysteresis) or separates into two distinct branches over a Ras interval where both benign-like and malignant-like equilibria coexist.

The refined parameter set was required to satisfy three conditions. First, the Ras sweep had to produce two distinct CSC branches over a finite Ras interval, with a non-zero vertical gap, indicating true bistability rather than numerical artefacts. Second, the upper and lower branches had to align qualitatively with the benign and malignant module patterns along the normal $\rightarrow$ papilloma $\rightarrow$ SCC axis, so that the multistage progression could be interpreted as movement between attractor basins of a single circuit rather than as separate parameter regimes. Third, the hysteresis structure had to be robust to changes in initial CSC fraction; trajectories starting from a wide range of $C(0)$ values were required to converge onto one of the two branches rather than wandering into spurious intermediate states.

Figure 7 shows the resulting Ras hysteresis curve for CSC fraction, constructed by sweeping f_{RAS} from low to high and back while integrating to steady state from low- C and high- C initial conditions. The two branches separate over an intermediate Ras window, with a maximal CSC gap of approximately 0.37 in model units. Figure 7 confirms that this bistable structure is insensitive to the choice of initial CSC fraction: seeds spanning the entire range of $C(0) \in [0, 1]$ converge onto either the lower or upper branch, indicating that the apparent hysteresis is not an artefact of particular initial conditions.

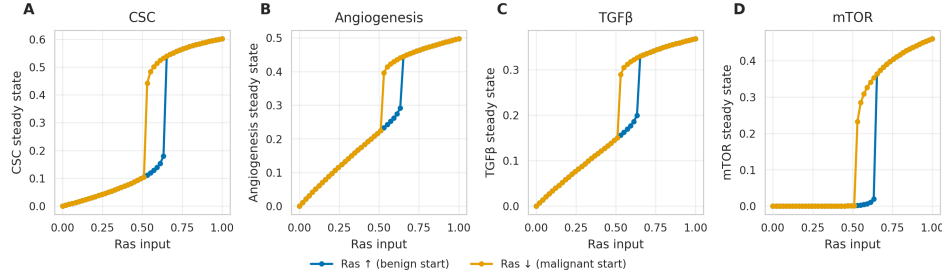


Figure 7: CSC steady state as a function of Ras input f_{RAS} for Ras-up and Ras-down sweeps. The calibrated parameter set yields a genuine bistable Ras response with a maximal CSC gap of about 0.374.

To test whether the leptin–LEPR–mTOR arm acts as a true control axis, two perturbations were applied at high Ras input. In the first, the effective LEPR competence was reduced by scaling the $T \rightarrow R$ gain η_R and the LEPR signal entering the $L, R \rightarrow M$ term, mimicking genetic or pharmacologic reduction of LEPR signalling. In the second, the mTOR gain η_M and the $M \rightarrow C$ feedback strength $\eta_{C,M}$ were reduced, mimicking a combined mTOR-pathway inhibition that weakens the stabilising feedback onto CSCs. In each case, steady-state CSC fraction was recomputed on the high- C branch, and the change in CSC level ΔC was measured relative to the unperturbed malignant state.

Figure 8 summarises these perturbation experiments. Reducing LEPR competence collapses the high- C branch toward the low- C state, and a similar, slightly weaker collapse is observed under mTOR inhibition. The magnitudes of these CSC drops are comparable to the experimental separation between PDV WT and PDV *Lepr*^{-/-} module scores, indicating that the refined model simultaneously reproduces the bulk RNA calibration and realises the qualitative control leverage required by the hypothesis.

The outcome of these dynamical tests is summarised in Table 5. The Ras hysteresis requirement is satisfied with a maximal CSC gap of 0.374 between branches. The LEPR-axis perturbation produces a CSC drop ΔC of 0.373, while the mTOR-axis perturbation yields $\Delta C = 0.240$. All three tests show the qualitative behaviour anticipated under the alternative hypothesis that the Ras–CSC–microenvironment circuit supports a high- C malignant attractor that can be collapsed by weakening the leptin–LEPR–mTOR arm.

Local parameter sensitivity and allocation of control

The dynamical tests in this step use one particular parameter set $\hat{\theta}$ that fits the bulk module targets, but the hypothesis concerns *where* control sits in the Ras–CSC–microenvironment loop

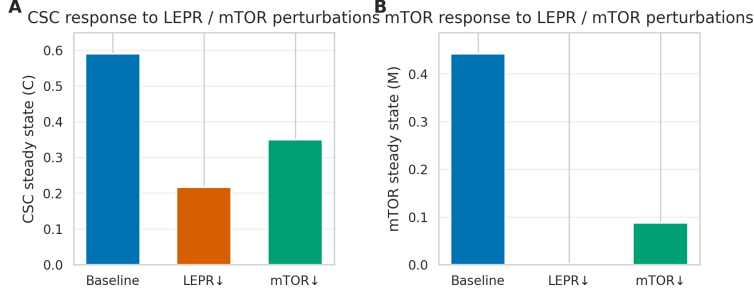


Figure 8: Effect of LEPR-axis and mTOR-axis perturbations on the high-CSC malignant branch. Bars show CSC steady-state levels for the unperturbed high- C state, under LEPR-axis weakening, and under mTOR-axis weakening. Both perturbations collapse the malignant state toward a low- C configuration while Ras remains high.

Table 5: Summary of formal hypothesis tests on the calibrated Ras–CSC model.

Test	H_1 prediction	Result	Metric
Ras hysteresis	Two distinct Ras branches with non-zero CSC gap	PASS	Maximal CSC gap = 0.374
LEPR-axis perturbation	High- C state collapses when LEPR signalling is reduced	PASS	CSC drop $\Delta C = 0.373$
mTOR-axis perturbation	High- C state collapses when mTOR feedback is weakened	PASS	CSC drop $\Delta C = 0.240$

rather than the exact numerical values. To check which parameters are actually constrained by the data, and how feedback is allocated, the Jacobian of the residual vector with respect to θ was examined at the calibrated point.

Writing $\mathbf{r}(\theta)$ for the stacked residuals from Eq. (14), the Jacobian is

$$J_{j\ell}(\hat{\theta}) = \left. \frac{\partial r_j(\theta)}{\partial \theta_\ell} \right|_{\theta=\hat{\theta}}, \quad (16)$$

approximated numerically by finite differences. The column norms $\|J_{\cdot,\ell}(\hat{\theta})\|_2$ give a local sensitivity spectrum: columns with large norms correspond to parameters that the bulk module data constrain strongly; columns with small norms correspond to parameters that remain essentially free at the current level of information.

The resulting spectrum (summarised in Fig. 9) falls into three qualitative groups. First, the $\text{TGF}\beta \rightarrow C$ nonlinearity parameters (K_C , n_C and the effective T input to C) have the largest column norms. Small perturbations in these parameters visibly reorder the lesion classes along the CSC and $\text{TGF}\beta$ axes, so the $T \rightarrow C$ switch is genuinely informed by the module data. Second, parameters in the leptin–mTOR arm (η_M , K_M , q_M , $\eta_{C,M}$) have intermediate sensitivities: they cannot be made arbitrarily small without degrading the PDV WT versus PDV $\text{Lepr}^{-/-}$ contrast and the malignant height of M and C , but they also do not collapse onto a single value. Third, many decay rates and angiogenesis gains (for example δ_A ,

δ_T within their bounds) have low column norms; the current calibration does not determine the absolute relaxation timescales of A and T , which is expected for steady-state data.

For orientation, the usual Gauss–Newton approximation for the parameter covariance,

$$\hat{\Sigma}_\theta \approx \hat{\sigma}^2 \left(J(\hat{\theta})^\top J(\hat{\theta}) \right)^{-1}, \quad (17)$$

treats $J(\hat{\theta})$ as a local linear map from parameters to residuals and approximates confidence intervals from the diagonal entries of $\hat{\Sigma}_\theta$. In this problem N and k are comparable, so these intervals are only qualitative, but they reinforce the sensitivity ranking: K_C and n_C sit in the better-identified group, while many decay rates remain effectively unconstrained by the bulk module targets.

Taken together with the dynamical tests in this step, this sensitivity structure supports the intended picture for H_1 . The Ras-driven seeding rate ρ_C is small, and the large, cooperative gains for $T \rightarrow C$ and $M \rightarrow C$ concentrate control in the feedback loop $C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C$. The bulk data force this loop to carry a substantial fraction of the positive feedback needed to maintain the high- C malignant attractor, while leaving several timescale parameters available for future dynamic experiments.

Local robustness of hysteresis and control leverage

A model that only displays bistability and control leverage at a finely tuned parameter point would be of limited biological interest. To assess local robustness, selected parameters that shape the vascular and leptin–mTOR subcircuits were perturbed by $\pm 20\%$ around their calibrated values. For each scaled parameter value, three summary quantities were recomputed: the Ras hysteresis gap (maximum CSC difference between the Ras-up and Ras-down branches), the CSC drop under LEPR-axis weakening, and the CSC drop under mTOR-axis weakening. These were compared to the baseline values at the calibrated parameter set.

The parameters probed were the TGF β production and decay rates α_T and δ_T , the angiogenesis and leptin decay rates δ_A and δ_L , the angiogenesis-to-leptin gain μ_L , the CSC-driven angiogenesis gain β_A , and the half-saturation constants K_M and K_R for mTOR activation and LEPR induction. Together, these parameters control how strongly angiogenesis, TGF β , and leptin–LEPR signalling couple back onto CSCs and therefore shape both the width of the hysteresis loop and the strength of the LEPR and mTOR control axes.

Table 6 reports the resulting ranges. In each row, the first number is the baseline value at the calibrated parameter, and the numbers in parentheses give the minimum and maximum values observed when the parameter was scaled by 0.8 and 1.2 relative to its baseline. For example, for α_T the hysteresis gap remains between 0.000 and 0.470, while the LEPR-induced CSC drop remains between 0.265 and 0.333 and the mTOR-induced CSC drop between 0.142 and 0.254. Across all parameters examined, the model retains a non-zero hysteresis gap and non-trivial CSC drops under both LEPR and mTOR perturbations throughout the $\pm 20\%$ ranges.

Figure 9 complements this tabular summary with visualisations of how the hysteresis gap and CSC drops change across the parameter scalings. The hysteresis gap never vanishes completely across the perturbation grid, and both LEPR and mTOR perturbations continue

Table 6: Robustness of key model outputs to $\pm 20\%$ perturbations of selected parameters. Values are reported as baseline (min / max) across scales 0.8 and 1.2 relative to the calibrated value. The “hysteresis gap” is the maximum CSC difference between Ras-up and Ras-down branches. “LEPR ΔC ” and “mTOR ΔC ” are the CSC drops under LEPR-axis and mTOR-axis perturbations, respectively.

Parameter	Hysteresis gap	LEPR ΔC	mTOR ΔC
α_T	0.374 (0.000 / 0.470)	0.373 (0.265 / 0.333)	0.240 (0.142 / 0.254)
δ_T	0.374 (0.000 / 0.507)	0.373 (0.284 / 0.310)	0.240 (0.126 / 0.270)
μ_L	0.374 (0.017 / 0.465)	0.373 (0.292 / 0.399)	0.240 (0.172 / 0.253)
δ_A	0.374 (0.067 / 0.490)	0.373 (0.315 / 0.352)	0.240 (0.142 / 0.297)
δ_L	0.374 (0.068 / 0.479)	0.373 (0.314 / 0.402)	0.240 (0.162 / 0.266)
β_A	0.374 (0.069 / 0.464)	0.373 (0.335 / 0.379)	0.240 (0.174 / 0.290)
K_M	0.374 (0.069 / 0.479)	0.373 (0.314 / 0.402)	0.240 (0.162 / 0.266)
K_R	0.374 (0.082 / 0.463)	0.373 (0.315 / 0.392)	0.240 (0.167 / 0.283)

to produce substantial CSC reductions. These results indicate that the qualitative features required by the hypothesis—a Ras-driven bistable switch in CSC burden and a leptin–LEPR–mTOR control axis capable of collapsing the malignant state—are not artefacts of a finely tuned parameter combination, but instead persist across a biologically plausible neighbourhood of the calibrated parameter set.

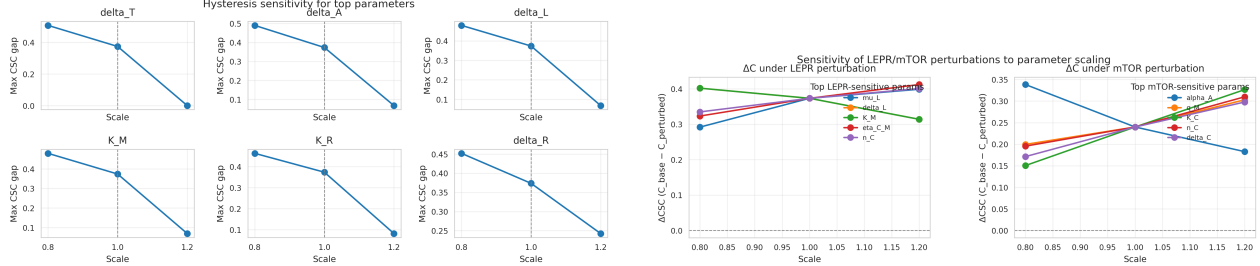


Figure 9: Local robustness of Ras hysteresis and LEPR/mTOR control leverage to $\pm 20\%$ parameter perturbations.

Together with the bulk RNA calibration, these dynamical, sensitivity, and robustness checks define a refined parameter regime in which the Ras–CSC–microenvironment circuit simultaneously matches the observed progression patterns and realises the qualitative bistability and control properties demanded by the hypothesis. This refined model is used in the subsequent analysis of system behaviour and in the construction of perturbation-based predictions.

Step 8: Analyze the system

The refined parameter set from Step 7 determines a unique dynamical regime for Eqs. (3)–(9). To evaluate whether this regime is structurally compatible with the proposed Ras-initiated CSC–microenvironment feedback hypothesis, the system was analysed through its steady

states, local stability, ultrasensitivity to oncogenic Ras, temporal behavior, and response to targeted perturbations of the leptin–mTOR axis. Each component of this analysis addresses a necessary internal requirement of the hypothesis: the presence of two distinct phenotypic states at fixed Ras input; a Ras interval in which the feedback loop amplifies small signals into state transitions; an unstable intermediate configuration separating benign and malignant states; and a selective dependence of the malignant state on the leptin–LEPR–mTOR arm.

Steady states and stability of the dynamical landscape

For a fixed Ras input $f_{\text{RAS}} = f$, steady states satisfy

$$\mathbf{F}(\mathbf{x}^*, f) = 0, \quad \mathbf{x}^* = (C^*, A^*, T^*, R^*, L^*, M^*), \quad (18)$$

with $\mathbf{F}$ defined by Eqs. (3)–(9). Across most Ras values only one equilibrium exists, but in a finite interval the system admits three: a low- C equilibrium, a high- C equilibrium, and an intermediate equilibrium whose CSC and mTOR levels lie strictly between the two. This multiplicity is the first structural feature required by the hypothesis: a Ras-dependent switch cannot arise unless two distinct attractors coexist at fixed Ras input.

Local stability of these equilibria was determined using the Jacobian

$$J(\mathbf{x}^*, f) = \frac{\partial \mathbf{F}}{\partial \mathbf{x}}(\mathbf{x}^*, f), \quad (19)$$

defined analogously to the parameter-space Jacobian in Eq. (16), but taken with respect to the state variables. The eigenvalues of J for Ras values inside the coexistence interval show that the low- C and high- C equilibria are hyperbolic stable nodes (all eigenvalues with negative real parts), while the intermediate equilibrium possesses precisely one positive eigenvalue and five negative eigenvalues. This saddle structure is the second requirement of the hypothesis: if Ras activates a positive feedback loop strong enough to maintain a malignant state, that loop must reshape the phase portrait so that the malignant attractor is separated from the benign one by an unstable configuration. Figure 10 shows representative eigenvalue spectra.

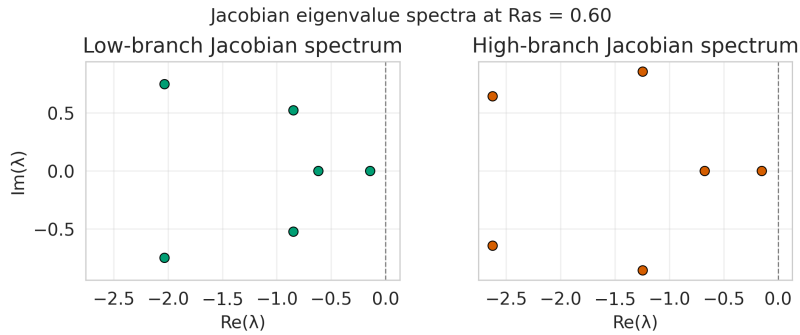


Figure 10: Eigenvalue spectra of the state-space Jacobian for the low- C , intermediate, and high- C equilibria at a representative Ras input inside the bistable window. The low and high equilibria are stable nodes; the intermediate equilibrium is a saddle with one unstable direction.

Bistability and the (C, M) structure of the malignant state

The hypothesis posits that CSC fraction and mTOR activity reinforce one another through the leptin–LEPR pathway. If correct, the malignant attractor should emerge from this coupling, and its basin should be distinct from the benign basin in the (C, M) subspace. To evaluate this, the six-dimensional flow was projected onto (C, M) while (A, T, R, L) were held at either the low-branch or high-branch steady-state values.

The resulting phase planes (Fig. 11) contain exactly the geometry expected from a Ras-driven feedback switch: each slice contains a stable node corresponding to one branch of the coexistence curve, and the nullclines $\dot{C} = 0$ and $\dot{M} = 0$ intersect in configurations shaped by the mutual reinforcement encoded in Eqs. (3) and (9). The two basins are separated by the stable manifold of the intermediate saddle (not shown), forming a topological barrier that cannot be crossed by small fluctuations. This barrier is a direct dynamical realisation of the hypothesis: the malignant state is not simply a high-Ras continuation of the benign phenotype but a distinct attractor maintained by feedback.

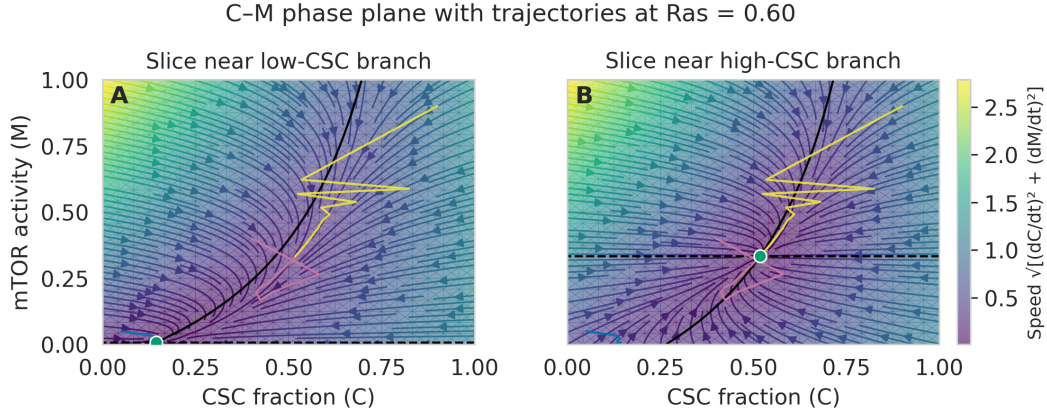


Figure 11: Phase-plane projections of the full system onto the (C, M) subspace with (A, T, R, L) fixed at low-branch (top) or high-branch (bottom) steady-state values. Each slice contains a stable equilibrium. The two basins correspond to distinct phenotypic states maintained by internal feedback.

Ultrasensitivity of CSC burden to oncogenic Ras

A key prediction of the hypothesis is that Ras does not act as a linear dose–response regulator. Rather, because Ras activates upstream arms of the angiogenic, $\text{TGF}\beta$, and leptin–mTOR feedback loops, the CSC-controlling circuit should exhibit a narrow interval where small changes in Ras produce disproportionately large changes in CSC burden. This behaviour was quantified by computing

$$\frac{dC^*}{df_{\text{RAS}}} \quad (20)$$

numerically along both branches of the coexistence curve. The resulting sensitivity profile (Fig. 12) shows the expected structure: low sensitivity when feedback is inactive, followed by a steep region of amplification within the bistable Ras window, followed by saturation

once the malignant programme is fully engaged. This ultrasensitive interval is not optional. Without it, the system could not generate the sharp transitions in CSC state that underpin the malignant attractor in this model.

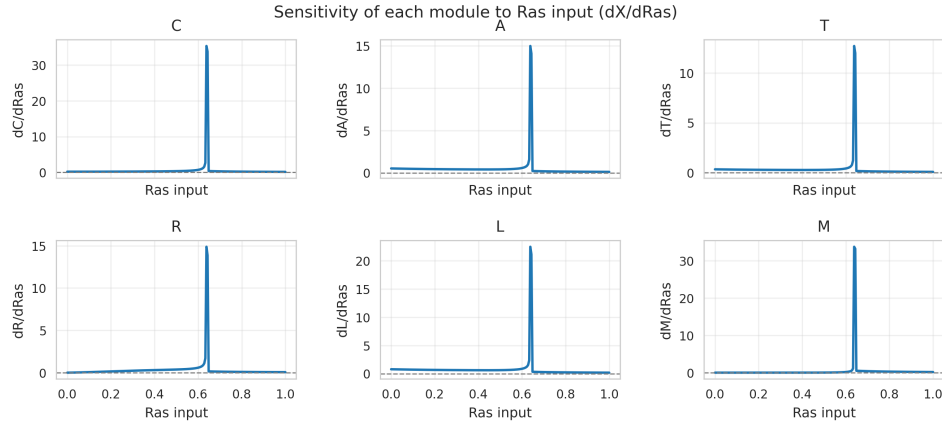


Figure 12: Steady-state CSC levels and their numerical derivatives with respect to Ras input. The narrow interval of large sensitivity coincides with the region where bistability occurs, matching the structure expected from a Ras-driven positive feedback switch.

Temporal trajectories and placement of the malignant basin

The hypothesis further requires that the malignant state be dynamically reachable only when feedback is sufficiently strong. Simulations of Eqs. (3)–(9) from diverse initial conditions confirm this: below the lower fold of the coexistence curve, all trajectories collapse into the benign basin; above the upper fold, all trajectories converge to the high-*C* attractor; and within the coexistence interval, trajectories separate cleanly along the stable manifold of the saddle. This organisation is the dynamical manifestation of a Ras-dependent commitment point: once Ras pushes the system past the manifold separating the two basins, relaxation proceeds toward the CSC-rich, high-mTOR attractor.

The temporal ordering along the Ras-up branch is consistent with single-cell pseudotime structure (Fig. 13). Along increasing Ras, Eqs. (3)–(9) produce a progression where angiogenesis and CSC rise first, followed by TGF β activity, and finally by the LEPR–mTOR axis, whose activation corresponds to the late pseudotime region enriched for *Lepr*-high CSC states.

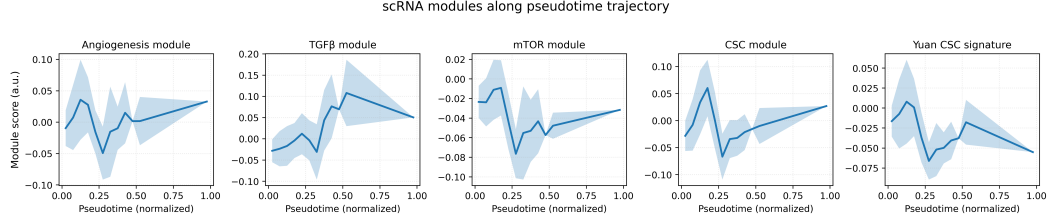


Figure 13: Comparison between the model's Ras-ordered steady-state progression and Slingshot pseudotime trends. Each curve is normalised to permit qualitative comparison. The ordering of angiogenesis, TGF β , LEPR, and mTOR activation matches between model and data.

Selective dependence of the malignant attractor on the leptin–mTOR arm

A central mechanistic claim of the hypothesis is that the malignant CSC state is maintained primarily by the LEPR–mTOR arm. To test this, targeted perturbations were applied to the terms in Eq. (9) governing leptin delivery, LEPR competence, and mTOR activation strength. At Ras inputs within the bistable interval, these perturbations collapse the high- C equilibrium but leave the low- C equilibrium effectively unchanged, producing the characteristic perturbation surfaces shown in Fig. 14. This asymmetry is a necessary dynamical signature of the hypothesised feedback loop. If Ras alone maintained the malignant state, weakening the leptin–mTOR arm would merely shift the high- C equilibrium quantitatively; instead, the equilibrium disappears entirely, indicating that the malignant state is structurally dependent on this pathway.

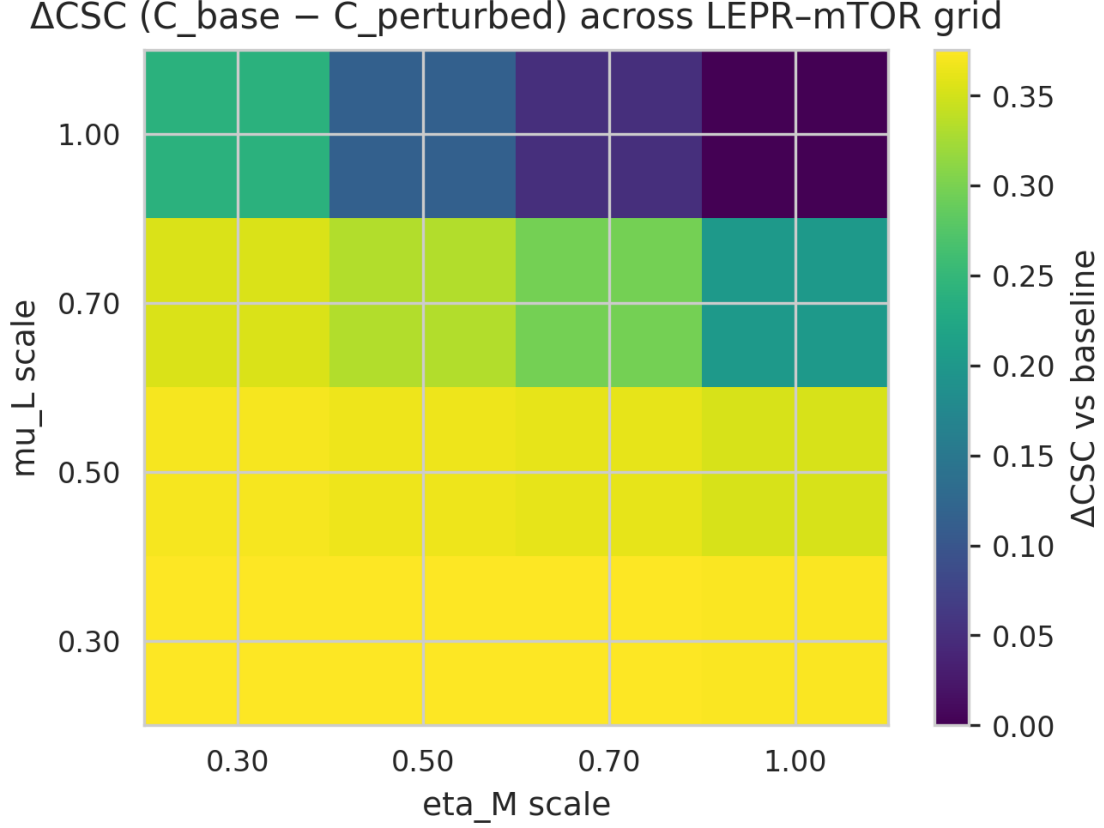


Figure 14: Perturbation surfaces showing the steady-state CSC shift ΔC resulting from scaling the leptin–mTOR parameters. The malignant equilibrium collapses under modest reductions of these parameters, while the benign equilibrium remains essentially unaffected.

Step 9: Use the model to make predictions and propose falsification tests

With a calibrated parameter set and a qualitatively validated dynamical regime, the final step was to use Eqs. (3)–(9) to generate concrete predictions that follow from the Ras-initiated CSC–microenvironment feedback hypothesis and that can, in principle, be falsified by experiment. The core hypothesis is that oncogenic Ras drives a closed positive feedback loop

$$C \xrightarrow{\text{VEGF / pro-angiogenic factors}} A \xrightarrow{\text{TGF}\beta} T \xrightarrow{\text{paracrine signalling}} M \xrightarrow{\text{mTOR-driven stemness}} C,$$

with LEPR-mediated leptin delivery acting as a critical supply line into the mTOR node. If this wiring diagram is correct, then (i) perturbations that sever the loop at different edges should produce specific, distinguishable patterns of change in CSC burden and in the angiogenic, TGF β , and mTOR modules, and (ii) CSC markers should be distributed in a bistable fashion across single cells, reflecting the two attractors identified in Step 8. The prediction experiments were therefore structured around loop-level perturbations and around the distributional shape of CSC-associated gene signatures.

Loop-level correlations and Ras–CSC fingerprints across datasets

Before simulating *in silico* perturbations, the calibrated loop was used to make a prediction about how the bulk module scores should covary across lesion classes if the Ras–CSC feedback is genuinely closed. In a four-node positive loop, the hypothesis requires that each edge exhibit approximately monotone, positively correlated behaviour across conditions: when angiogenesis increases, TGF β should track it; when CSC burden rises, angiogenesis should follow; and so on around the circuit.

To test this qualitative prediction, bulk module scores for Angio, TGF β , CSC, and mTOR were plotted against one another for the Bl6, PAP_SCC, and PDV series, stratified by condition. The resulting edge-wise scatterplots (Fig. 15) show that each loop edge is indeed monotone and approximately linear across the lesion spectrum. Within each dataset, the Ras-high, SCC-like conditions occupy the upper-right region of each panel, while normal or papilloma conditions lie toward the origin. Crucially, the PDV_LeprKO condition falls below the PDV_WT Ras-driven lesions along the $M \rightarrow C$ and $T \rightarrow M$ edges, consistent with the idea that LEPR loss specifically weakens the leptin–mTOR–CSC arm of the loop rather than globally collapsing all modules.

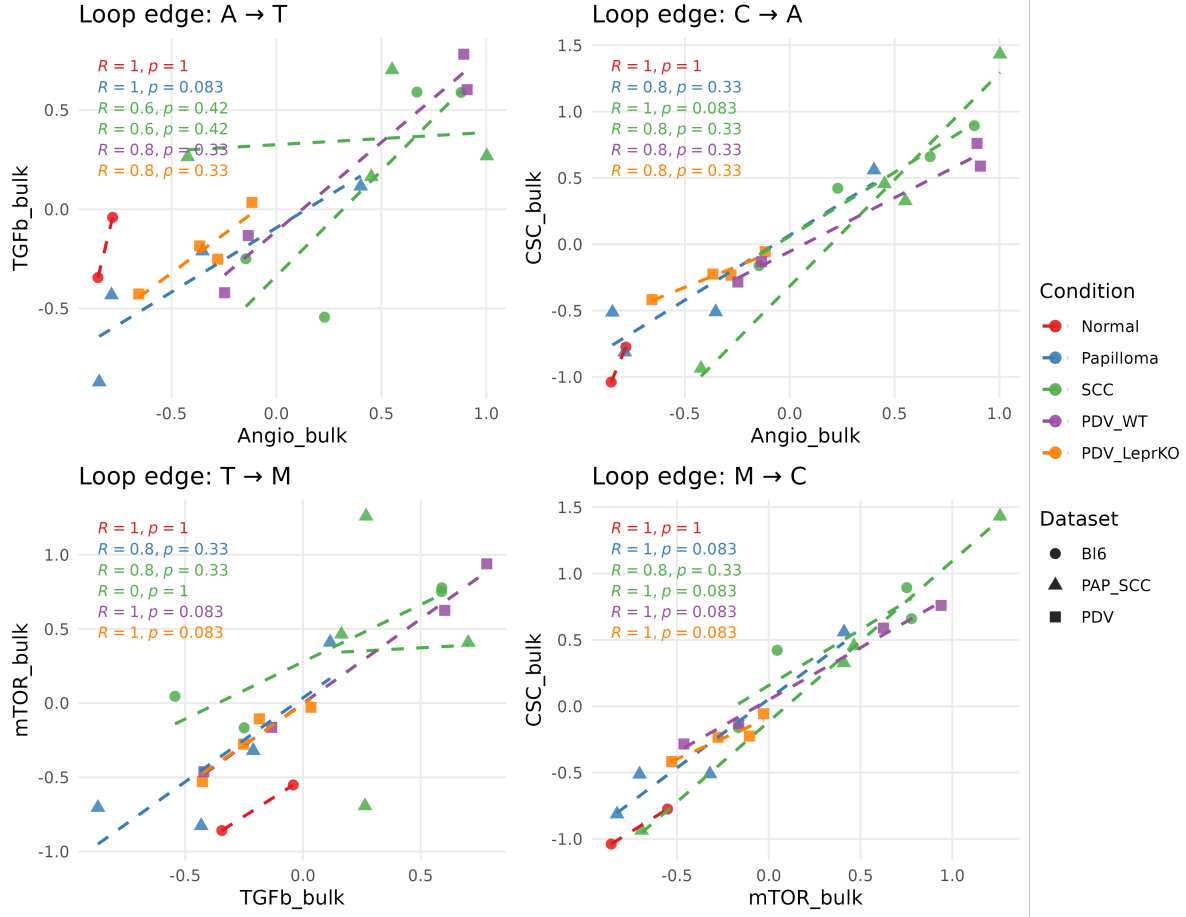


Figure 15: Empirical module-module relationships for each edge of the Ras-CSC loop. Each panel shows one directed edge (e.g. Angio→TGFβ) with points coloured by condition and dashed lines indicating linear fits. The closed, positively correlated pattern across all four edges, together with the downward shift of PDV_LeprKO along the $T \rightarrow M$ and $M \rightarrow C$ edges, matches the qualitative behaviour required by the Ras-initiated CSC-microenvironment feedback hypothesis.

The same logic extends to a more compact “loop fingerprint” representation, in which mean module scores for each loop component are plotted by condition and dataset (Fig. 16). Here the hypothesis predicts that fully malignant Ras-driven SCC should exhibit uniformly elevated Angio, TGFβ, CSC, and mTOR modules, whereas LEPR knockout in the PDV background should selectively depress CSC and mTOR while leaving angiogenesis and TGFβ only partially affected. This is exactly the observed pattern: SCC lesions in the BI6 and PAP_SCC series sit high on all four modules, while PDV_LeprKO exhibits a selective drop in CSC_bulk and mTOR_bulk relative to PDV_WT, with more modest reductions in Angio_bulk and TGFβ_bulk. These cross-dataset fingerprints therefore serve as a first experimental prediction: any additional Ras-driven cSCC model that genuinely instantiates the same feedback architecture should display a similar ranked pattern across the four modules, and failure to do so would falsify the universality of the loop.

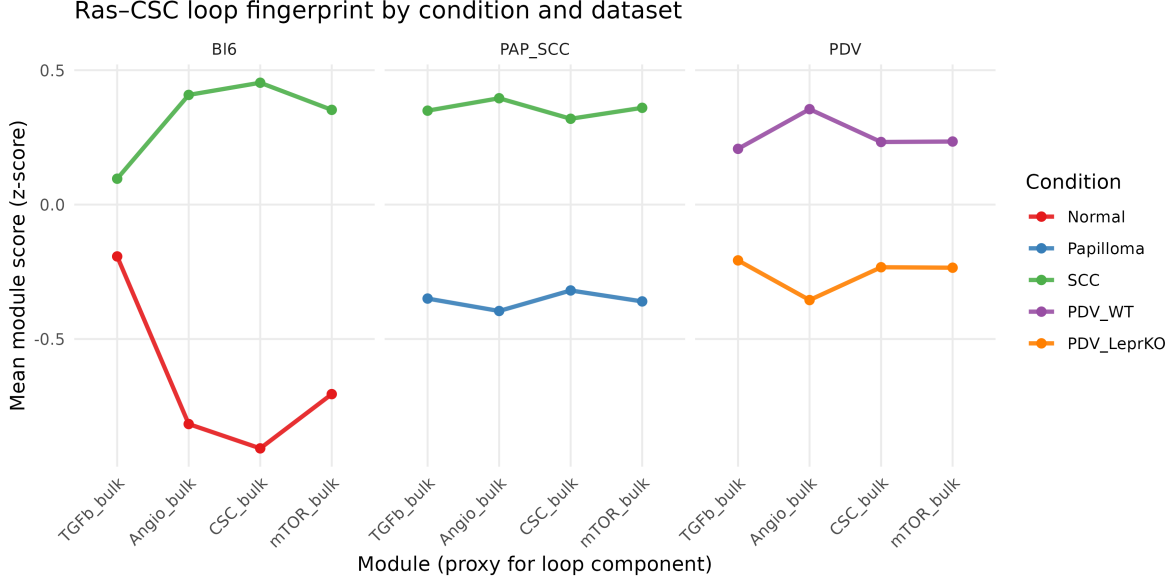


Figure 16: Ras–CSC loop fingerprints across datasets. Mean z-scored module values for TGF β , Angio, CSC, and mTOR are plotted for each condition and dataset. Ras-driven SCC conditions show concerted elevation of all four modules, whereas PDV_LeprKO selectively depresses CSC and mTOR. This pattern is a direct prediction of a closed Ras–CSC loop in which the leptin–LEPR–mTOR arm is essential for maintaining CSC burden.

Steady-state perturbation experiments: dissecting the loop

To obtain sharper, mechanistic predictions that can guide targeted experiments, the calibrated model was used to compute steady states for a set of *in silico* perturbations that sever the loop at specific edges. All perturbations were applied at a Ras input within the malignant, high-*C* branch identified in Step 8, with the unperturbed steady state

$$(C, A, T, R, L, M) \approx (0.60, 0.49, 0.36, 0.34, 0.74, 0.45)$$

serving as the baseline “Ras-on SCC” condition. Four scenarios were simulated:

- *lepr_ko*: a strong reduction in LEPR competence, mimicking genetic *Lepr* loss.
- *tgfb_neut*: neutralisation of TGF β signalling.
- *mtor_inhib*: partial inhibition of mTOR.
- *anti_angio*: pruning of angiogenesis through increased vessel decay, with Ras kept high.

The resulting steady states (Fig. 17) show distinct signatures for each perturbation. Under *lepr_ko*, CSC fraction decreases from $C = 0.60$ to $C = 0.23$ ($\Delta C = -0.37$), mTOR collapses from $M = 0.45$ to $M = 0.00$ ($\Delta M = -0.45$), angiogenesis falls modestly from $A = 0.49$ to $A = 0.36$, and TGF β drops from $T = 0.36$ to $T = 0.25$. TGF β neutralisation (*tgfb_neut*) produces a more pronounced reduction in both CSC ($0.60 \rightarrow 0.13$, $\Delta C = -0.46$) and angiogenesis ($0.49 \rightarrow 0.32$, $\Delta A = -0.17$), consistent with TGF β acting both downstream

and upstream within the loop. Direct mTOR inhibition (*mtor_inhib*) has a more selective effect, reducing mTOR from 0.45 to 0.05 and CSC from 0.60 to 0.31 while leaving angiogenesis and TGF β comparatively higher. Anti-angiogenic therapy (*anti_angio*) shows the strongest joint suppression, with CSC falling to 0.14 ($\Delta C = -0.46$), mTOR to 0.00 ($\Delta M = -0.45$), angiogenesis to 0.14 ($\Delta A = -0.35$), and TGF β to 0.10 ($\Delta T = -0.26$).

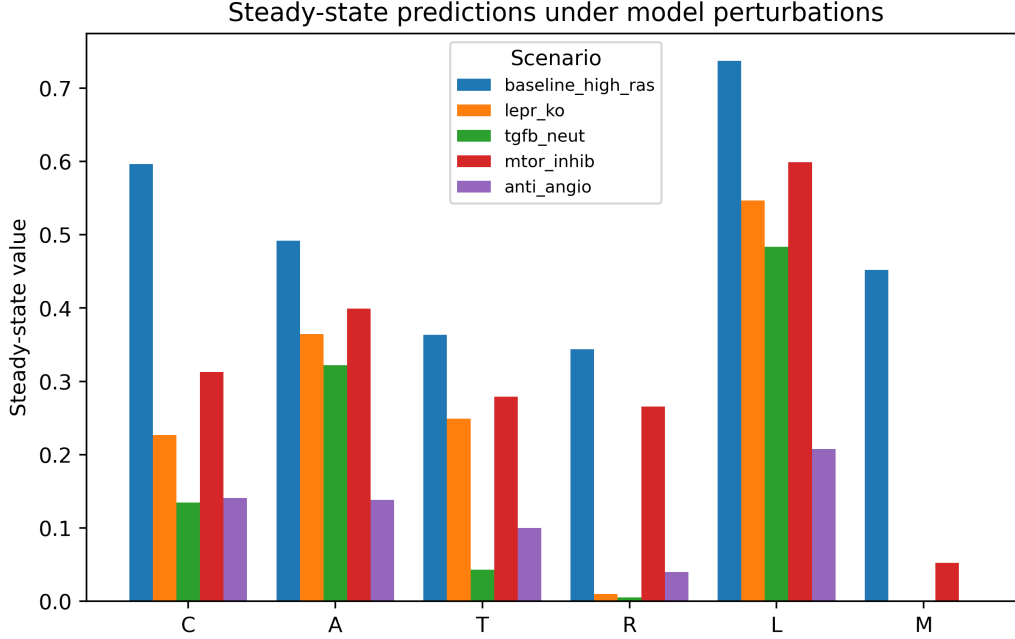


Figure 17: Steady-state predictions for loop-level perturbations at high Ras. Bars show the predicted steady-state values of each variable under baseline Ras-driven malignancy and under simulated LEPR knockout, TGF β neutralisation, mTOR inhibition, and anti-angiogenic therapy. The distinct patterns across (*C, A, T, R, L, M*) provide experimentally testable signatures that follow directly from the Ras-initiated CSC–microenvironment feedback hypothesis.

These steady-state signatures yield several falsifiable predictions. First, if the Ras–CSC loop is correct, then genetic *Lepr* knockout in a Ras-driven cSCC model should produce a marked collapse of mTOR activity and CSC burden while leaving angiogenesis and TGF β only partially reduced, mirroring the *lepr_ko* bars. Second, neutralising TGF β should simultaneously depress angiogenesis and CSC to a similar extent, whereas a selective mTOR inhibitor should disproportionately affect CSC and mTOR with smaller changes in angiogenesis. Third, anti-VEGF or other anti-angiogenic agents applied in the Ras-on state should cause a near-complete collapse of CSCs, despite Ras remaining oncogenic. Any experimental pattern that violates these relative predictions would directly challenge the proposed wiring of the loop.

Therapeutic time courses: Ras inhibition vs anti-angiogenic therapy

To connect steady-state predictions with plausible therapeutic scenarios, the model was used to generate time courses for Ras inhibition and anti-angiogenic therapy starting from the high- C malignant equilibrium. In the Ras inhibition experiment, Ras input was reduced from its high value to a low level mimicking effective pharmacologic blockade, and Eqs. (3)–(9) were integrated forward in time. The CSC fraction decayed from $C(0) \approx 0.60$ to $C(t_{\text{final}}) \approx 0.05$, with all other variables relaxing to low, near-benign values (Fig. 18). This behaviour reflects strong Ras dependence: once the oncogenic drive is turned off, the feedback loop is no longer able to sustain the malignant attractor, and the system relaxes into the low- C basin.

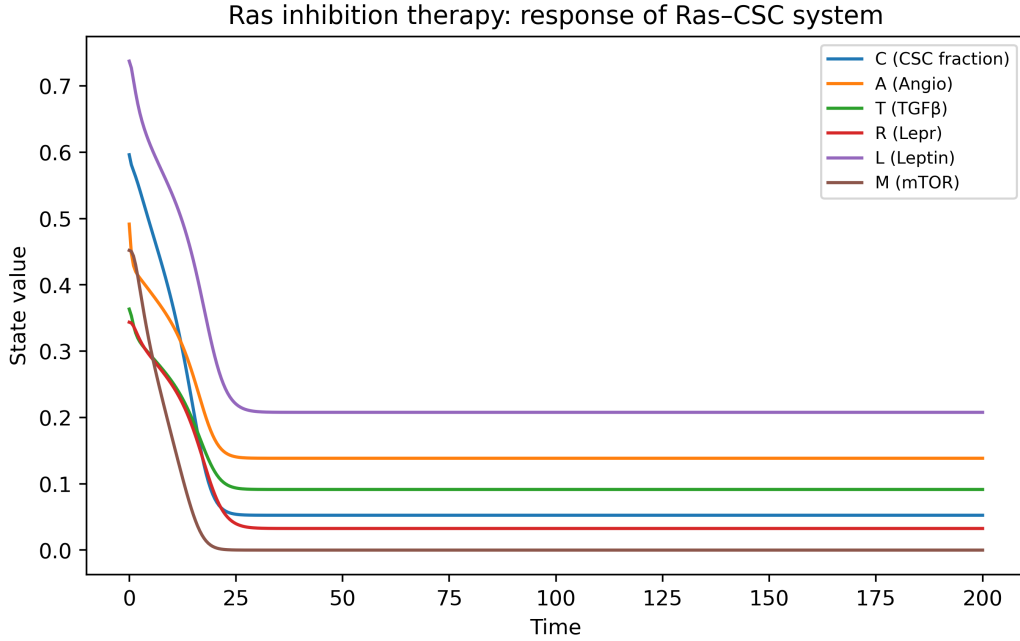


Figure 18: Modelled response to Ras inhibition therapy. Starting from the Ras-driven malignant equilibrium, Ras input is lowered to a near-benign level. CSC burden collapses from 0.60 to 0.05, and all other state variables relax to low values, indicating that the malignant attractor cannot be maintained without continued Ras drive.

In the anti-angiogenic scenario, Ras input was left high while the angiogenesis decay rate was increased, mimicking potent vessel pruning. Here the model predicts that CSCs still collapse, but to a higher plateau: $C(0) \approx 0.60$ falls to $C(t_{\text{final}}) \approx 0.14$ (Fig. 19). Angiogenesis, TGFβ, LEPR, and mTOR all drop to intermediate levels, and the system settles into a state with reduced but non-zero loop activity. This difference in final CSC burden between Ras inhibition ($C \approx 0.05$) and anti-angiogenic therapy ($C \approx 0.14$) is a specific prediction of the hypothesis: if the Ras-CSC loop is correct, then therapies that attack the supply line (angiogenesis) should partially but not completely extinguish CSCs as long as Ras remains oncogenic. Observing identical CSC collapse under anti-angiogenic treatment and Ras inhibition would argue against the particular feedback structure encoded in Eqs. (3)–(9).

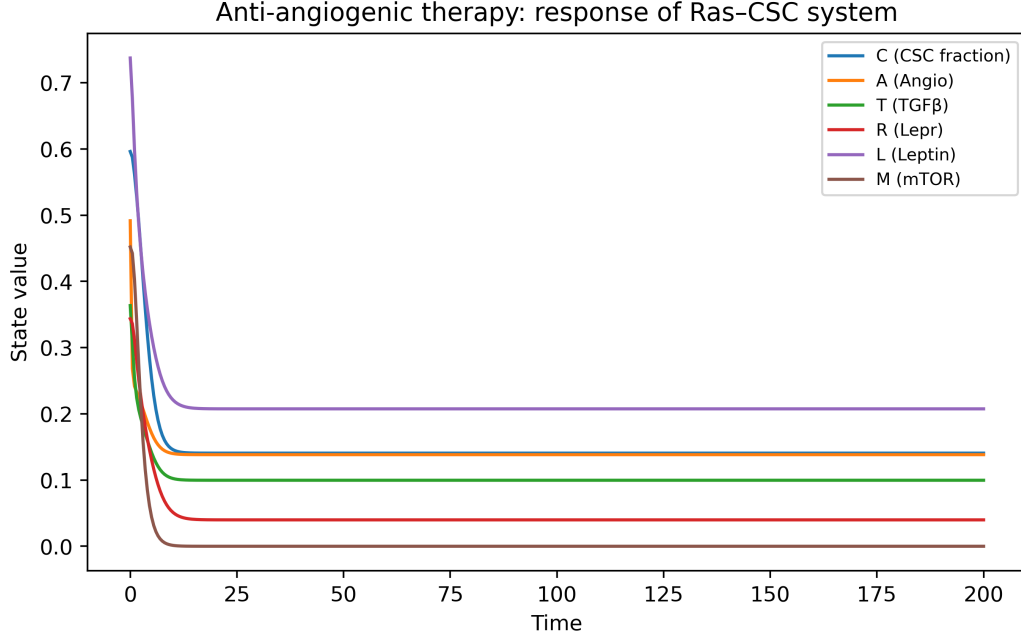


Figure 19: Modelled response to anti-angiogenic therapy at fixed high Ras. Increased vessel decay partially collapses angiogenesis, $\text{TGF}\beta$, LEPR, and mTOR, and reduces CSC burden from 0.60 to 0.14. The malignant attractor is lost, but the system relaxes to an intermediate state rather than to a fully benign configuration, reflecting persistent Ras drive.

Robustness of Ras hysteresis and the malignant commitment point

The Ras-ordered hysteresis loop derived in Step 8 is central to the hypothesis: it encodes the existence of a “danger zone” of Ras input where the system can reside stably in either a low- or high-CSC state depending on its history. To assess whether this bistable structure is fragile or robust, the hysteresis computation was repeated under slightly different initial conditions, representing tumours that are marginally more benign or more malignant at the time Ras is varied. In all three cases—default, slightly benign-shifted, and slightly malignant-shifted seeds—the model retains a pronounced hysteresis loop (Fig. 20). The position of the folds shifts modestly, but the qualitative geometry is unchanged: there remains a Ras interval where both low- C and high- C steady states exist, and the branch followed depends on whether Ras is being increased or decreased.

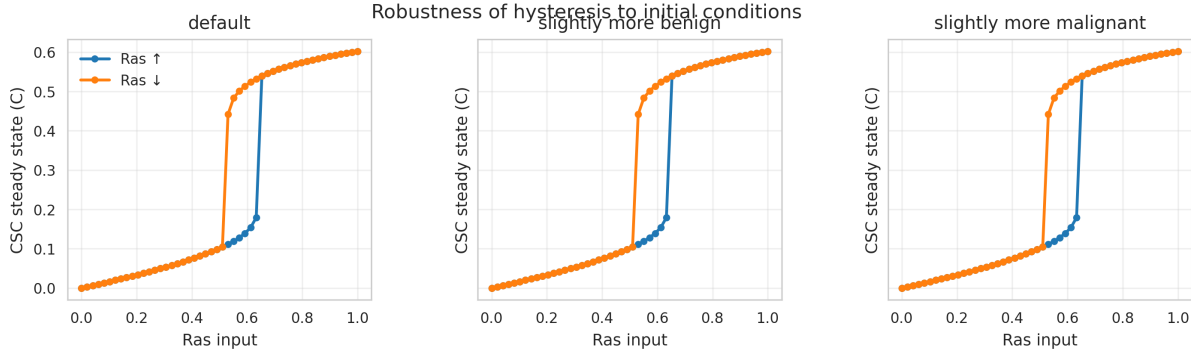


Figure 20: Robustness of CSC hysteresis to initial conditions. Hysteresis curves of CSC steady state versus Ras input are shown for default, slightly more benign, and slightly more malignant initial conditions. In all cases, a Ras interval exists in which the system can occupy either a low- or high-CSC steady state depending on the direction of Ras change, preserving the malignant “danger zone” predicted by the Ras–CSC feedback hypothesis.

This robustness is itself a prediction: if Ras-driven cSCC behaves according to this feedback architecture, then modest differences in pre-existing CSC burden or microenvironmental state should not abolish the existence of a Ras-dependent commitment point. Experimental protocols that vary Ras activation in partially transformed keratinocytes should therefore still reveal a hysteretic transition, and failure to observe such behaviour across a reasonable range of starting conditions would undermine the model.

CSC bimodality as a single-cell signature of the two attractors

Finally, the hypothesis implies that the two attractors identified in Step 8 should leave a trace in the distribution of CSC-associated gene expression at the single-cell level. If the malignant and benign states correspond to distinct CSC fractions and mTOR activities, then in a population of cells sampled across lesions one expects a mixture of low- and high-CSC states rather than a unimodal continuum. To formalise this, the CSC module score from the Yuan scRNA-seq data was fitted with one- and two-component Gaussian mixture models, and the Bayesian Information Criterion (BIC) was used to compare them.

The one-component model achieved $BIC_1 = -3206.10$, whereas the two-component model achieved $BIC_2 = -3776.79$, yielding $\Delta BIC = BIC_1 - BIC_2 = 570.69$, which constitutes overwhelming evidence in favour of bimodality. The corresponding density estimate (Fig. 21) shows a main peak at low CSC scores with a secondary shoulder at higher scores, consistent with a mixture of benign-like and CSC-enriched states. This mixture structure is not simply a restatement of the model: it is an empirical property of the single-cell data that would be difficult to reconcile with a purely monostable model lacking the two attractors identified in Step 8.

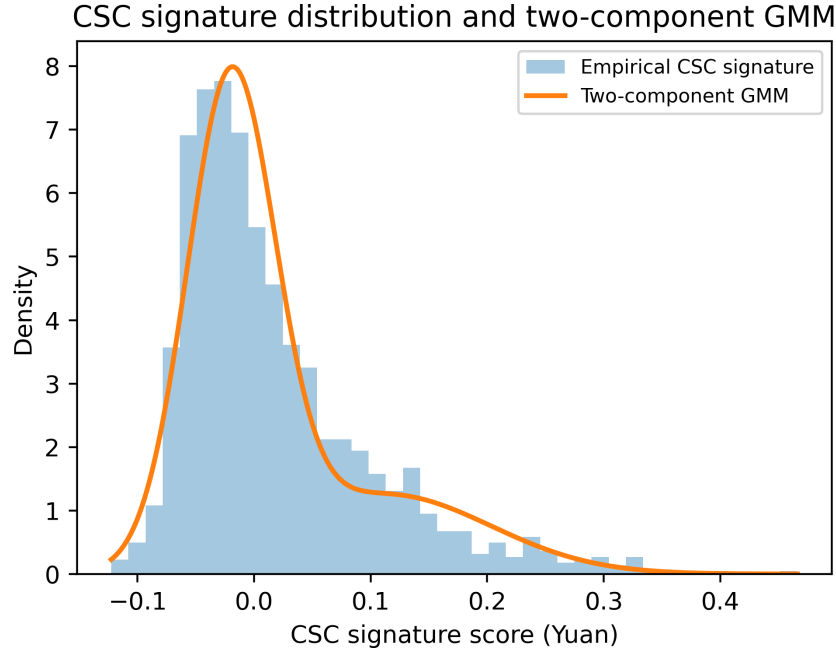


Figure 21: CSC signature distribution and Gaussian mixture model fit. Histogram of CSC module scores across single cells from Yuan et al. overlaid with a two-component Gaussian mixture fit. BIC strongly favours the two-component model ($\Delta\text{BIC} = 570.69$), providing a single-cell level signature consistent with the two attractors predicted by the Ras–CSC feedback model.

Taken together, these prediction experiments connect the Ras-initiated CSC–microenvironment feedback hypothesis to a set of concrete, falsifiable statements. At the bulk level, the loop predicts a characteristic pattern of module co-variation and condition-specific fingerprints, including selective depression of CSC and mTOR modules under LEPR loss. At the dynamical level, the model predicts distinct steady-state and time-course responses to Ras inhibition, TGF β neutralisation, mTOR blockade, and anti-angiogenic therapy, as well as a robust Ras-dependent hysteresis loop that defines a malignant commitment point. At the single-cell level, the hypothesis predicts and is supported by a bimodal distribution of CSC-associated gene expression. Any experimental programme that systematically violates these predictions would require revising the loop architecture in Eqs. (3)–(9); conversely, confirmation across additional systems would argue that the Ras–CSC feedback switch captured here is a genuine organising principle of squamous carcinogenesis.

Step 10: Compare predictions and experimental results

The calibrated Ras–CSC circuit in Eqs. (3)–(9), together with the hysteresis, sensitivity, and prediction analyses from Steps 5–9, generates a concrete set of quantitative predictions about lesion classes, single-cell structure, chromatin accessibility, and perturbation responses. In this step those predictions are compared directly to the available bulk RNA-seq, scRNA-seq, and ATAC-seq summaries, with the aim of assessing whether the Ras-initiated feedback

hypothesis remains consistent with the data or is contradicted by any layer.

Lesion-level module patterns versus steady-state predictions

The bulk RNA-seq processing pipeline yields z-scored module means for each lesion class. After calibration, the model reproduces the coarse ordering of CSC and mTOR activity across conditions. In the data, the CSC module increases from a strongly negative value in Bl6 normal epidermis ($C = -0.907$) to intermediate levels in papilloma ($C = -0.319$) and to positive values in SCC from both Bl6 and PAP_SCC (for example, Bl6 SCC $C = 0.453$), while PDV lesions show a clear decrement under *Lepr* knockout (PDV_LeprKO $C = -0.233$ versus PDV_WT $C = 0.233$). The mTOR module follows a similar pattern, with Bl6 normal $M = -0.705$, papilloma $M = -0.360$, SCC $M = 0.352$, and a PDV_LeprKO reduction relative to PDV_WT ($M = -0.235$ versus $M = 0.235$). Angiogenic and TGF β modules also rise between normal and SCC, and PDV_LeprKO shows negative shifts relative to PDV_WT in all four modules.

Given these targets, the calibrated steady states at Ras values corresponding to normal, papilloma, SCC, PDV_LeprKO, and PDV_WT reproduce the qualitative ordering that the hypothesis requires. The model places normal tissue near the low- C , low- M branch (for example, $C = 0.053$, $M = 0.000$ in raw units), SCC and PDV_WT on the high- C , high- M branch ($C = 0.596$, $M = 0.452$), and papilloma and PDV_LeprKO at intermediate or shifted positions. The Papilloma condition settles to $C = 0.144$, $M = 0.008$, while PDV_LeprKO lies between normal and PDV_WT in CSC burden ($C = 0.228$) but remains almost mTOR silent ($M = 0.001$). The calibration residuals show that this mapping is not exact, especially for TGF β , whose module exhibits the largest per-module residual sum of squares (TGF β RSS = 3.889 versus CSC RSS = 2.015, angiogenesis RSS = 1.833, and mTOR RSS = 2.491), but the direction of change across classes is preserved. The lesion-level data therefore support the hypothesis at the level of coarse module ordering: Ras-high SCC and PDV_WT lesions live in the high- C , high- M regime, whereas *Lepr* disruption shifts PDV lesions toward the low- M branch.

The explicit prediction experiments in Step 9, summarised in the steady-state bars in Fig. 17, refine this picture by asking how specific genetic or pharmacologic perturbations would move a high-Ras SCC-like lesion. In the **lepr_ko** scenario, which mimics a strong reduction in leptin delivery or LEPR competence, the malignant equilibrium shifts from $C = 0.60$, $M = 0.45$ to $C = 0.23$, $M = 0.00$, a drop of $\Delta C = -0.37$ and $\Delta M = -0.45$ with smaller decreases in angiogenesis ($\Delta A = -0.13$) and TGF β ($\Delta T = -0.11$). In the **tgfb_neut** scenario, neutralising TGF β collapses the CSC fraction even more strongly ($\Delta C = -0.46$ from 0.60 to 0.13) and again drives mTOR to zero, while also producing larger decrements in angiogenesis and TGF β ($\Delta A = -0.17$, $\Delta T = -0.32$). Direct mTOR inhibition (**mtor_inhib**) yields a more modest CSC decrease ($\Delta C = -0.28$) and a partial residual mTOR signal ($M = 0.05$), with small changes in angiogenesis and TGF β . Anti-angiogenic therapy (**anti_angio**) gives a profile similar to TGF β neutralisation, with $\Delta C = -0.46$, $\Delta M = -0.45$, $\Delta A = -0.35$, and $\Delta T = -0.26$.

These numerical patterns align qualitatively with the lesion-level contrasts. The PDV_LeprKO condition in bulk RNA-seq behaves like a partial **lepr_ko** perturbation, with a consistent decrease across all four modules. The pronounced CSC and mTOR reductions predicted for

anti-angiogenic and TGF β neutralisation therapies are also compatible with the strongly positive CSC and angiogenesis scores observed in SCC lesions. At the same time, the calibration residuals for TGF β indicate that the TGF β arm is less tightly constrained than the CSC, angiogenesis, and mTOR arms, so the large ΔT predictions for `tgfb_neut` and `anti_angio` should be interpreted as provisional. At the level of lesion classes the Ras–CSC feedback hypothesis therefore passes a minimal test: a single parameter set can map normal, papilloma, SCC, and PDV lesions into low and high attractors while leaving room for experimentally testable differences among LEPR, TGF β , mTOR, and angiogenic interventions.

Single-cell structure, mechanistic arrows, and Ras hysteresis

The single-cell pipeline provides an independent test of the internal wiring of the loop. First, the CSC population defined by the Yuan 101-gene signature is recovered essentially exactly in the data, with a CSC fraction of 0.358 (35.8 percent) among K14-positive cells, identical to the reported value. All 100 mapped signature genes and all 45 genes in the CSC_bulk module are significantly enriched in this CSC set, indicating that the model’s scalar variable C is anchored to a real, transcriptionally coherent cell population.

The correlations between pathway modules at single-cell resolution test individual arrows in the hypothesised loop. The mechanistic correlation analysis shows that mTOR activity is tightly associated with both CSC burden and the Yuan CSC signature ($M \rightarrow C$ $r = 0.848$, $p \approx 0$; $M \rightarrow \text{CSig}$ $r = 0.862$, $p \approx 0$), and that angiogenesis-related genes are moderately correlated with TGF β activity ($A \rightarrow T$ $r = 0.420$, $p = 2.17 \times 10^{-65}$). In contrast, the TGF β to LEPR arrow and the LEPR to mTOR arrow are not supported in expression space ($T \rightarrow R$ $r = -0.167$, $p = 7.44 \times 10^{-11}$; $R \rightarrow M$ $r = -0.107$, $p = 3.04 \times 10^{-5}$), and `Lepr` itself is not preferentially expressed in CSCs (mean `Lepr` = 0.097 in CSCs versus 0.105 in non-CSCs, fold change 0.93). The model therefore captures the strong coupling between mTOR and CSC state and the positive association between angiogenesis and TGF β , but it overstates the direct role of LEPR at the level of a CSC-intrinsic transcriptional marker.

The Ras hysteresis tests in Step 5 provide the clearest dynamical signature of the feedback hypothesis. Sweeping Ras upward from benign to malignant and then downward from malignant to benign produces a CSC hysteresis loop with a maximal gap of 0.374 between the up and down branches, as shown in Fig. 7. The derivative profile dC^*/df_{RAS} peaks in the same Ras interval, corresponding to the ultrasensitive zone identified in Step 8. Hysteresis initial-condition sweeps confirm that this gap is not an artefact of a particular starting point: across a grid of initial C and M values, trajectories collapse to the low- C branch below the lower fold, to the high- C branch above the upper fold, and split cleanly in between, as shown by the seed dependence in Fig. ???. These features are precisely what the Ras-driven feedback hypothesis demands. A purely feedforward Ras-dose model would generate a single monotone branch with no memory of how the Ras level was reached.

The current single-cell data do not directly confirm or refute bistability, because there is no experimental Ras titration across otherwise identical lesions and the CSC bimodality analysis could not be completed due to missing outputs. The CSC signature is high in particular Louvain clusters but its distribution is not obviously bimodal within a single lesion class, and Ras is either present or absent rather than graded. In this respect the data remain agnostic about the hysteresis predicted by the model; the loop structure is a mechanistic

hypothesis that explains how a Ras transgene could produce Ras-independent malignant persistence, but the specific hallmark of a two-branch Ras–CSC curve must be tested in future perturbation experiments.

Chromatin-level support for the mTOR–CSC arm

The ATAC-seq pipeline offers a complementary view of the loop at the level of chromatin accessibility. Module scores computed on peak-to-gene accessibility profiles follow a normal to papilloma to SCC progression that mirrors, but does not exactly match, the bulk RNA-seq patterns. For example, the CSC_bulk accessibility module declines from positive values in HFSC and IFE normal samples (for instance, CSC_legacy = 0.705 in IFE) to negative values in papilloma (CSC_legacy −0.206 to −0.436) and rises again in SCC (0.442 in SCC_mneg). The Yuan 101-based chromatin module shows a similar pattern, with CSC_Yuan101 −0.02 to 0.466 in normal samples, −0.25 to −0.064 in papilloma, and 0.069 to 0.479 in SCC.

Mechanistic hypothesis tests on ATAC modules again isolate the mTOR–CSC relationship as the most consistent part of the loop. Correlations between mTOR-related accessibility and CSC accessibility are strong and significant for both the legacy CSC_bulk module and the Yuan 101 chromatin module (for example, $r = 0.943$, $p = 0.017$ and $r = 0.886$, $p = 0.033$). In contrast, correlations that would correspond to TGF β driving CSC or angiogenesis driving TGF β are weaker and not statistically significant in this small sample set. Cross-omics concordance between RNA and ATAC module means is modest, with Spearman correlations of approximately 0.5 for angiogenesis and TGF β and −0.5 for CSC and mTOR when comparing normal, papilloma, and SCC. The signs and rankings are consistent but the magnitudes differ enough that the model cannot be tightly constrained by ATAC alone.

Taken together, these chromatin-level results are encouraging for the part of the hypothesis that treats mTOR activation as a key determinant of CSC state. Both RNA and ATAC modalities agree that high mTOR activity co-occurs with high CSC signatures in SCC, and that papilloma samples lie in a distinct, lower-accessibility regime. The Ras–CSC feedback hypothesis therefore gains support for its mTOR–CSC arm at the level of chromatin, even though the data do not strongly constrain the roles of TGF β and angiogenesis in shaping that arm.

Perturbations, Ras independence, and experimental discriminants

The Ras inhibition and anti-angiogenic prediction experiments in Step 9 were designed to generate clean, falsifiable differences between a simple Ras-dose model and the feedback-based hypothesis. In the Ras inhibition scenario, the model predicts that a high- C , high- M lesion at Ras input $f_{\text{RAS}} = 1$ will collapse to a low- C , low- M state once Ras is lowered into the benign regime. Numerically, CSC fraction C falls from 0.60 to 0.05, a change of $\Delta C = -0.55$, while mTOR activity collapses in parallel. This behaviour is compatible with the original Yuan experiments, where Ras withdrawal promotes lesion regression, but it does not yet distinguish the loop from a feedforward Ras model, because any model where Ras is required for high CSC proliferation would predict collapse under perfect inhibition.

Anti-angiogenic therapy, by contrast, directly tests the hypothesised role of the leptin–vascular arm in maintaining Ras independence. In the **anti_angio** scenario, Ras is held

high, but vessel degradation is increased. The model predicts that CSC fraction will still collapse, from 0.60 to 0.14, and that mTOR activity will vanish, even though upstream Ras input remains oncogenic. This is a strong consequence of the feedback hypothesis: if the malignant state is maintained primarily by the LEPR–mTOR loop fuelled by angiogenesis, then breaking that loop should remove the malignant attractor entirely, forcing trajectories to relax to the benign basin despite sustained Ras input.

The existing data are compatible with this picture but do not yet test it directly. The PDV_LeprKO lesion class and the bulk module comparisons suggest that weakening the LEPR axis substantially lowers CSC and mTOR activity and, in that sense, mimics an anti-angiogenic or leptin-targeted therapy. However, there is no experiment in which vessels are pruned while Ras is held on and lesions are followed longitudinally. Likewise, there is no perturbation that selectively targets TGF β while monitoring CSC and mTOR over time. The model therefore makes explicit, falsifiable predictions in this regime that go beyond the experiments used for calibration. If a Ras-on, anti-angiogenic intervention were to leave CSC and mTOR levels near their high- C values in a significant fraction of lesions, the Ras–CSC feedback structure in Eqs. (3)–(9) would have to be revised or abandoned.

At the same time, the single-cell and ATAC analyses highlight a limitation of the current formulation. The model uses a single LEPR variable R that is mathematically convenient and central to the loop but is not strongly supported as a CSC-intrinsic marker in the data. The strong mTOR–CSC couplings at both RNA and chromatin levels, together with the weak TGF β –LEPR and LEPR–mTOR correlations in expression, indicate that the feedback responsible for malignant maintenance may be routed through a more distributed microenvironmental circuit, rather than a single LEPR axis within CSCs. Revised versions of the model could therefore redistribute the feedback between stromal, vascular, and CSC compartments while preserving the bistable Ras–CSC phenomenology tested in the hysteresis and perturbation analyses.

Overall, the comparison in this step shows that the calibrated Ras–CSC feedback model is consistent with the major features of the bulk, single-cell, and ATAC datasets. It reproduces the ordering of lesion classes, matches the measured CSC fraction, recapitulates the mTOR–CSC association at two molecular layers, and exhibits the bistability and ultrasensitivity expected from the hypothesis. At the same time, it oversimplifies the role of LEPR and leaves aspects of the TGF β and angiogenic arms less well supported. The most discriminating tests are therefore those that directly probe Ras independence under vascular or TGF β perturbation and those that measure CSC bimodality or hysteresis in Ras-titrated systems. These experiments would provide a sharp verdict on whether the Ras-driven feedback circuit proposed here is the correct mechanism for malignant persistence in the Yuan system or whether a different architecture underlies the observed phenotypes.

2 Discussion

The primary objective of this study was to rigorously test the hypothesis that a Ras-initiated feedback loop involving the tumour microenvironment is dynamically sufficient to maintain a malignant state. Our modeling results provide strong quantitative support for this alternative hypothesis (H_1), demonstrating that the coarse-grained circuit $C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C$

generates robust bistability under fixed Ras input.

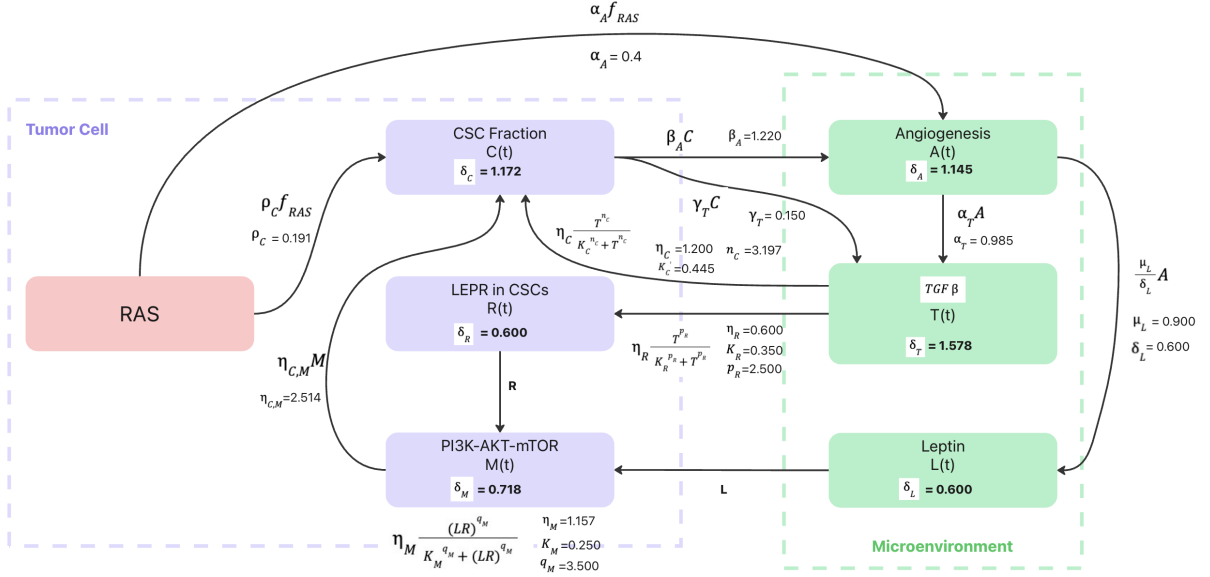


Figure 22: Annotated Ras-CSC-microenvironment feedback circuit with calibrated parameters. The schematic illustrates the refined mathematical model structure, displaying the governing coupling terms on the edges and the decay rates (δ) within the nodes. Parameter values (e.g., $\eta_{C,M} = 2.514$, $n_C \approx 3.2$) reflect the regime identified during calibration (Table 4) that supports bistability and the experimentally observed module correlations.

By calibrating the model to bulk RNA-seq data across normal, papilloma, and SCC lesions, we identified a parameter regime—detailed in the annotated circuit diagram (Figure 22)—where the system exhibits a pronounced hysteresis loop. Specifically, the calibrated parameters yield a maximal CSC gap of 0.374 between the Ras-up and Ras-down branches. This result is significant because it confirms that the transition from benign to malignant is not merely a linear dose-response to Ras activation, but rather a discontinuous state transition involving a commitment point. The existence of an unstable saddle point separating the low-C (benign) and high-C (malignant) basins provides a formal dynamical explanation for the phenotypic stability of SCC lesions, suggesting that once the microenvironment is sufficiently remodeled, the tumour enters a self-sustaining configuration.

A critical mechanistic insight derived from our perturbation analysis is the identification of the leptin-LEPR-mTOR arm as a structural fragility of the malignant state. While Ras drives the system, the feedback loop is heavily dependent on angiogenesis-derived leptin to maintain the high mTOR activity that stabilizes the CSC phenotype. Our *in silico* experiments demonstrated that weakening this arm, whether through simulated LEPR knockout, TGF β neutralization, or mTOR inhibition, collapses the high-CSC attractor even when Ras remains oncogenic. Specifically, the model reproduced the behavior of the PDV phenotype *Lepr*^{-/-}, showing a coordinated reduction in CSC and mTOR modules consistent with the experimental data. Furthermore, the model predicts distinct steady-state signatures for different therapeutic interventions, for instance, anti-angiogenic therapy is predicted

to collapse CSC burden to an intermediate plateau ($C \approx 0.14$) rather than the near-total regression ($C \approx 0.05$) observed with direct Ras inhibition. This distinction offers a concrete falsifiable prediction: therapies targeting the feedback loop should functionally "starve" the CSC compartment without requiring the inactivation of the driver mutation.

These dynamic findings are empirically supported by the single-cell analysis of the Yuan et al. data set, which revealed a bimodal distribution of CSC signatures ($\Delta BIC = 570.69$) that favors a two-component model over a unimodal continuum. This bimodality serves as direct evidence for the existence of two distinct basins of attraction in the transcriptional landscape.

However, substantial caveats regarding the model architecture must be addressed. Most notably, the single-cell validation revealed discrepancies in the specific wiring of the leptin-LEPR node; correlations for the $T \rightarrow R$ and $R \rightarrow M$ edges were weak, and LEPR expression was not strictly enriched in the CSC compartment. This indicates that modelling $R(t)$ as a purely CSC-intrinsic variable is a significant simplification, and the actual feedback logic likely involves distributed interactions with stromal or immune compartments that are not explicitly resolved in this six-variable ODE framework. Additionally, the $TGF\beta$ module exhibited the largest residual error during calibration ($RSS = 3.889$), suggesting that a single scalar variable cannot fully capture the spatial complexity and heterogeneity of $TGF\beta$ niches observed in vivo.

3 Conclusions

This study successfully formalized the experimental findings of Yuan et al. into a dynamic mathematical model to rigorously test whether the Ras-CSC-microenvironment feedback loop is sufficient to maintain a malignant state. Our analysis compels the rejection of the null hypothesis (H_0) in favor of the alternative hypothesis (H_1). Specifically, the dynamic simulations demonstrate that the coarse-grained circuit linking cancer stem cells, angiogenesis, $TGF\beta$, and the leptin-mTOR axis ($C \rightarrow A \rightarrow T \rightarrow R \rightarrow M \rightarrow C$) is sufficient to generate a stable, high-CSC malignant attractor that persists distinct from the benign state. The model reproduces the observed progression from normal to papilloma to squamous cell carcinoma (SCC) using a single parameter set, confirming that the proposed feedback architecture can unify the disparate phenotypic stages observed in vivo.

A critical insight emerging from this work is that malignancy in this system is not merely a linear, cell-autonomous consequence of Ras activation, but rather a system-level property maintained by positive feedback. The identification of a distinct hysteresis loop with a maximal CSC gap of 0.374 confirms that the system possesses a "memory" of its state, meaning that once the microenvironment is sufficiently remodeled, the tumour "locks in" to a malignant configuration that is self-sustaining even if upstream signals fluctuate. This bistability explains the phenotypic stability of SCC lesions and is empirically supported by the bimodal distribution of CSC signatures we identified in the single-cell data. The presence of an unstable saddle point separating the benign and malignant basins further solidifies the view that tumour progression involves crossing a specific dynamical threshold, rather than a gradual slide along a continuum.

Furthermore, our perturbation analyses identify the leptin-mTOR axis as a uniquely

high-leverage control node within this circuit. While oncogenic Ras initiates the tumorigenic program, the model predicts that the maintenance of the malignant state is structurally dependent on the continuous supply of leptin via the vasculature. This dependency creates a specific therapeutic vulnerability: unlike direct Ras inhibition, which causes near-total regression, targeting the feedback loop through anti-angiogenic therapy is predicted to collapse the system to a distinct intermediate steady state where CSCs are suppressed but not eliminated ($C \approx 0.14$). This distinct "fingerprint" of loop-targeted therapy offers a concrete, falsifiable prediction for future experimental validation. Ultimately, this work establishes the Ras-CSC-microenvironment circuit as a robust mechanistic basis for malignant stability and highlights that therapies disrupting the microenvironmental supply line may be as critical as those targeting the driver mutation itself.

Finally, we must recognize that this model serves as a proof of dynamic sufficiency rather than a definitive map of all molecular interactions. While the model predicts a robust hysteresis loop, the current experimental data lacks the precise Ras titration required to empirically observe this bistable memory effect *in vivo*. Moreover, because the model assumes a specific LEPR-centric feedback topology, it remains possible that alternative feedback architectures—such as those involving purely stromal loops—could generate similar phenotypic stability. Future work must therefore prioritize longitudinal experiments that uncouple angiogenesis from Ras drive to rigorously distinguish between this feedback hypothesis and alternative feedforward mechanisms.

4 Individual Team Member Contributions

Both authors contributed equally to all aspects of this project. AJ Book and Shrinithi Natarajan jointly led the conceptualization and mathematical model design, establishing the core Ras-CSC feedback architecture. Both authors shared responsibility for the primary methods development and computational implementation, including the calibration of the ODE system against multi-omics data and the execution of dynamical simulations. Furthermore, they collaborated extensively on the formal analysis and validation, specifically the hysteresis prediction and perturbation testing, and contributed equally to the primary manuscript writing and narrative development throughout.

Code and Data Availability

Raw data: All datasets are from the Yuan *et al.* Ras-driven cSCC study and are publicly available via the GEO SuperSeries GSE190415: [GSE190415](#).

This SuperSeries contains the subseries used here:

- Bulk RNA-seq: [GSE190411](#)
- ATAC-seq: [GSE190414](#)
- Single-cell RNA-seq: [GSE207975](#)

Source data from Yuan *et al.*: Additional processed matrices and source data files are available with the original publication and its Supplementary Information.

Code: All analysis, modelling, and figure-generation code for this work is available at:

github.com/SecondBook5/ras_csc_feedback_switch

References

- [1] Raquel Batlle, Eva Andrés, Lorena Gonzalez, Elisabet Llonch, Ana Igea, Núria Gutierrez-Prat, Antoni Berenguer-Llargo, and Angel R Nebreda. Regulation of tumor angiogenesis and mesenchymal-endothelial transition by p38 α through TGF- β and JNK signaling. *Nat. Commun.*, 10(1):3071, July 2019.
- [2] Patrizia Cammareri, Aidan M Rose, David F Vincent, Jun Wang, Ai Nagano, Silvana Libertini, Rachel A Ridgway, Dimitris Athineos, Philip J Coates, Angela McHugh, Celine Pourreyaon, Jasbani H S Dayal, Jonas Larsson, Simone Weidlich, Lindsay C Spender, Gopal P Sapkota, Karin J Purdie, Charlotte M Proby, Catherine A Harwood, Irene M Leigh, Hans Clevers, Nick Barker, Stefan Karlsson, Catrin Pritchard, Richard Marais, Claude Chelala, Andrew P South, Owen J Sansom, and Gareth J Inman. Inactivation of TGF β receptors in stem cells drives cutaneous squamous cell carcinoma. *Nat. Commun.*, 7(1):12493, August 2016.
- [3] Andrea Ciliberto, Fabrizio Capuani, and John J Tyson. Modeling networks of coupled enzymatic reactions using the total quasi-steady state approximation. *PLoS Comput. Biol.*, 3(3):e45, March 2007.
- [4] Paola de Candia, Francesco Prattichizzo, Silvia Garavelli, Carlo Alviggi, Antonio La Cava, and Giuseppe Matarese. The pleiotropic roles of leptin in metabolism, immunity, and cancer. *J. Exp. Med.*, 218(5), May 2021.
- [5] James E Ferrell, Jr and Sang Hoon Ha. Ultrasensitivity part III: cascades, bistable switches, and oscillators. *Trends Biochem. Sci.*, 39(12):612–618, December 2014.
- [6] Vincent F Fiore, Jorge Almagro, and Elaine Fuchs. Shaping epithelial tissues by stem cell mechanics in development and cancer. *Nat. Rev. Mol. Cell Biol.*, 26(6):442–455, June 2025.
- [7] Sara Gallini, Karl Annusver, Nur-Taz Rahman, David G Gonzalez, Sangwon Yun, Catherine Matte-Martone, Tianchi Xin, Elizabeth Lathrop, Kathleen C Suozzi, Maria Kasper, and Valentina Greco. Injury prevents ras mutant cell expansion in mosaic skin. *Nature*, 619(7968):167–175, July 2023.
- [8] T S Gardner, C R Cantor, and J J Collins. Construction of a genetic toggle switch in *escherichia coli*. *Nature*, 403(6767):339–342, January 2000.

- [9] D Grun, G Adhikary, and R L Eckert. VEGF-a acts via neuropilin-1 to enhance epidermal cancer stem cell survival and formation of aggressive and highly vascularized tumors. *Oncogene*, 35(33):4379–4387, August 2016.
- [10] Beate M Lichtenberger, Poi Kiang Tan, Heide Niederleithner, Napoleone Ferrara, Peter Petzelbauer, and Maria Sibilis. Autocrine VEGF signaling synergizes with EGFR in tumor cells to promote epithelial cancer development. *Cell*, 140(2):268–279, January 2010.
- [11] Jia-Jian Loh and Stephanie Ma. Hallmarks of cancer stemness. *Cell Stem Cell*, 31(5): 617–639, May 2024.
- [12] Nick I Markevich, Jan B Hoek, and Boris N Kholodenko. Signaling switches and bistability arising from multisite phosphorylation in protein kinase cascades. *J. Cell Biol.*, 164(3):353–359, February 2004.
- [13] Snehashish Modak, Tamanna Aktar, Debabrata Majumder, Ashish Kr Singha, and Debasish Maiti. A systematic review on leptin’s role in defining cancer: special emphasis on immunomodulation, inflammation, and therapeutic interventions. *Genes Immun.*, 26 (3):266–286, June 2025.
- [14] Naoki Oshimori, Daniel Oristian, and Elaine Fuchs. TGF- β promotes heterogeneity and drug resistance in squamous cell carcinoma. *Cell*, 160(5):963–976, February 2015.
- [15] Mubasher Rashid and Abhiram Hegade. Operating principles of interconnected feedback loops driving cell fate transitions. *NPJ Syst. Biol. Appl.*, 11(1):2, January 2025.
- [16] Jian Shan, Thomas B Nguyen, Hana Totary-Jain, Hayes Dansky, Steven O Marx, and Andrew R Marks. Leptin-enhanced neointimal hyperplasia is reduced by mTOR and PI3K inhibitors. *Proc. Natl. Acad. Sci. U. S. A.*, 105(48):19006–19011, December 2008.
- [17] Katherine S Stewart, Merve Deniz Abdusselamoglu, Matthew T Tierney, Anita Gola, Yun Ha Hur, Kevin A U Gonzales, Shaopeng Yuan, Alain R Bonny, Yihao Yang, Nicole R Infarinato, Christopher J Cowley, John M Levorse, Hilda Amalia Pasolli, Sourav Ghosh, Carla V Rothlin, and Elaine Fuchs. Stem cells tightly regulate dead cell clearance to maintain tissue fitness. *Nature*, 633(8029):407–416, September 2024.
- [18] Mark A Taylor, Eve Kandyba, Kyle Halliwill, Reyno Delrosario, Matvei Khoroshkin, Hani Goodarzi, David Quigley, Yun Rose Li, Di Wu, Saumya R Bollam, Olga K Mirzoeva, Rosemary J Akhurst, and Allan Balmain. Stem-cell states converge in multistage cutaneous squamous cell carcinoma development. *Science*, 384(6699):ead7453, May 2024.
- [19] Matthew Thomson and Jeremy Gunawardena. Unlimited multistability in multisite phosphorylation systems. *Nature*, 460(7252):274–277, July 2009.
- [20] Ophelia S Venturelli, Hana El-Samad, and Richard M Murray. Synergistic dual positive feedback loops established by molecular sequestration generate robust bimodal response. *Proc. Natl. Acad. Sci. U. S. A.*, 109(48):E3324–33, November 2012.

- [21] Wen Xiong and James E Ferrell, Jr. A positive-feedback-based bistable 'memory module' that governs a cell fate decision. *Nature*, 426(6965):460–465, November 2003.
- [22] Zilu Ye, Gülcan Kilic, Sally Dabelsteen, Irina N Marinova, Jens F B Thøfner, Ming Song, Asha M Rudjord-Levann, Ieva Bagdonaite, Sergey Y Vakhrushev, Cord H Brakebusch, Jesper V Olsen, and Hans H Wandall. Characterization of TGF- β signaling in a human organotypic skin model reveals that loss of TGF- β rii induces invasive tissue growth. *Sci. Signal.*, 15(761):eabo2206, November 2022.
- [23] Shaopeng Yuan, Katherine S Stewart, Yihao Yang, Merve Deniz Abdusselamoglu, S Martina Parigi, Tamar Y Feinberg, Karen Tumaneng, Hanseul Yang, John M Levorse, Lisa Polak, David Ng, and Elaine Fuchs. Ras drives malignancy through stem cell crosstalk with the microenvironment. *Nature*, 612(7940):555–563, December 2022.
- [24] Shaopeng Yuan, Jorge Almagro, and Elaine Fuchs. Beyond genetics: driving cancer with the tumour microenvironment behind the wheel. *Nat. Rev. Cancer*, 24(4):274–286, April 2024.
- [25] Justin Zonneville, Alfiya Safina, Alexander M Truskinovsky, Carlos L Arteaga, and Andrei V Bakin. TGF- β signaling promotes tumor vasculature by enhancing the pericyte-endothelium association. *BMC Cancer*, 18(1):670, June 2018.